# Contexts and pseudocontexts


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In the projection lattice of a finite-dimensional Hilbert space, an orthogonal basis—a *context*—is characterised by the property that its atomic probabilities sum to one on every state. We study a weakening of this condition in which two tuples of atoms induce the same probability sum on every state. Such pairs are called *pseudocontexts*. In finite dimension this is equivalent to two distinct rank-one decompositions of the same positive operator, placing the notion naturally among finite frame-theoretic constructions. Size $k = 1$ is trivial; size $k = 2$ already admits genuinely nonorthogonal examples over $\mathbb{C}$; and, in fact, examples exist for every $k \geq 2$. For $k \geq 3$, genuine pseudocontexts also exist over $\mathbb{R}$. We present the operator formulation, give a general explicit construction, analyse the spectral structure of the associated structure operator, illustrate the symmetric $k = 3$ case with a figure, and list the remaining classification problems.



Keywords: projection lattice, pseudocontext, rank-one decomposition, structure operator, quantum contextuality


## I. INTRODUCTION

Let $L$ be the lattice of projections in a Hilbert space of finite dimension $n \geq 2$. Its atoms (minimal non-zero elements) correspond to one-dimensional subspaces, represented by nonzero vectors. States are described by probability measures $P\colon L \to [0,1]$. We denote by $S(L)$ the set of all states on $L$.

Vectors $a_1, \ldots, a_n$ form a *context* if they compose an orthogonal basis. In this case (and only in this case) they satisfy

$$\forall P \in S(L): \quad \sum_{i=1}^{n} P(a_i) = 1. \tag{1}$$

The present note concerns weaker additive identifications obtained by comparing two tuples of atoms that are not required to be orthogonal. Such *pseudocontexts* are quantum-logical shadows of orthomodular-lattice constructions of Mayet–Navara–Rogalewicz [1] and have direct physical meaning: they encode state-independent empirical equivalences between families of generally noncommuting observables. It is then natural to ask for which sizes $k$ and over which scalar fields genuine pseudocontexts exist. The answer is simple: pseudocontexts already exist for $k = 2$ over $\mathbb{C}$, and, more generally, for every $k \geq 2$; for $k \geq 3$ they exist over $\mathbb{R}$ as well.

## II. PRELIMINARIES

Throughout, $H$ is a Hilbert space of finite dimension $n \geq 2$, and $L = L(H)$ is the orthomodular lattice of its closed subspaces. Its atoms correspond to one-dimensional subspaces, or equivalently to *rays* spanned by nonzero vectors. We identify an atom with any representative unit vector $a$ and write $P_a = |a\rangle\langle a|$ for the orthogonal projection onto the ray it spans. More generally, for nonzero $a$ we set $P_a = |a\rangle\langle a|/\langle a,a\rangle$, so that $P_a$ depends only on the ray $\mathbb{C}a$.

A *state* is a map $P\colon L \to [0,1]$ with $P(I) = 1$ that is finitely additive on orthogonal families. In dimension $n \geq 3$, Gleason's theorem [2] asserts that every state arises from a density operator $\rho$ via

$$P(a) = \operatorname{tr}(\rho P_a), \qquad \rho \geq 0, \quad \operatorname{tr}\rho = 1. \tag{2}$$

For $n = 2$, Gleason's theorem fails for the abstract definition of a state, and we adopt (2) as the definition: a state is the assignment $P_a \mapsto \operatorname{tr}(\rho P_a)$ for some density operator $\rho$. Under either reading, the key consequence we use repeatedly is that two self-adjoint operators that agree on every state are equal as operators (the density operators in finite dimension separate self-adjoint operators).

## III. CONTEXTS AND PSEUDOCONTEXTS

The operator form of (1) is elementary and is the starting point for everything that follows.

**Proposition 1** (Contexts)**.** *Unit vectors $a_1, \ldots, a_n \in H$ form an orthonormal basis if and only if*

$$\sum_{i=1}^{n} P_{a_i} = I. \tag{3}$$

*Proof.* The forward implication is immediate. For the converse, fix $j$ and evaluate (3) on the vector $a_j$:

$$1 = \langle a_j, I a_j\rangle = \sum_{i=1}^{n} |\langle a_i, a_j\rangle|^2.$$

The term with $i = j$ already contributes 1, so all others must vanish. Hence $\langle a_i, a_j\rangle = 0$ for $i \neq j$. As this holds for every $j$, the family is orthonormal, and being of size $n$ in an $n$-dimensional space it is an orthonormal basis. ☐


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Combined with (2), Proposition 1 recovers (1) as a state-theoretic characterisation of orthogonal bases.

**Definition 2** (Pseudocontext)**.** Let $k \geq 1$. A *pseudocontext of size* $k$ is a pair of $k$-tuples of atoms $(a_1,\dots,a_k)$, $(b_1,\dots,b_k)$ such that

(i) each tuple contains pairwise distinct atoms (i.e. no two of the $a_i$ are scalar multiples, and likewise for the $b_i$),

(ii) the two tuples are disjoint as sets of rays, so that $|\{a_1,\dots,a_k,b_1,\dots,b_k\}| = 2k$, and

(iii) $\forall P \in S(L): \quad \sum_{i=1}^{k} P(a_i) = \sum_{i=1}^{k} P(b_i).$

A pseudocontext is called *genuine* if neither $k$-tuple is a context.

*Remark* 3. Condition (i) excludes repeated atoms within a single tuple, and condition (ii) ensures the two tuples are truly distinct as collections of rays. Without condition (ii), every pseudocontext of size $k$ would yield pseudocontexts of every size $k' > k$ by adjoining $k'-k$ shared atoms; this trivial inflation can be carried out within the original ambient subspace. Condition (ii) rules out such padding and makes the size $k$ a genuine invariant of the construction.

By (2), condition (iii) of Definition 2 is equivalent to the operator identity

$$T := \sum_{i=1}^{k} P_{a_i} = \sum_{i=1}^{k} P_{b_i}, \tag{4}$$

which we call the *structure operator* of the pseudocontext. Thus pseudocontexts are precisely pairs of distinct unit-norm rank-one decompositions of a single positive operator. This places them naturally among finite frame-theoretic constructions: contexts are the special case in which the structure operator is the identity on the relevant subspace; in general, pseudocontexts are unit-norm rank-one frame decompositions of a common positive operator. If the common operator is proportional to the identity on its support, the family is a tight frame; after normalisation on that support one obtains a POVM.

*Remark* 4 (Connection to SIC-POVMs). When the common structure operator satisfies $T = \frac{k}{n} I$ (tight frame on all of $H$), each family $\{P_{a_1},\dots,P_{a_k}\}$ and $\{P_{b_1},\dots,P_{b_k}\}$ scaled by $n/k$ forms a symmetric informationally complete POVM (SIC-POVM) if $k = n^2$ and the frame vectors are equiangular [3]. Pseudocontexts therefore generalise the SIC-POVM notion by dropping both the tightness and equiangularity requirements.

## IV. THE SMALLEST CASES

### A. No pseudocontexts of size $k = 1$

**Proposition 5** (Size $k=1$)**.** *There is no pseudocontext of size $k=1$.*

*Proof.* The condition $P(a_1) = P(b_1)$ for all $P \in S(L)$ means that the two self-adjoint operators $P_{a_1}$ and $P_{b_1}$ have equal expectation in every density operator. In finite dimension, density operators separate self-adjoint operators, so $P_{a_1} = P_{b_1}$. This rank-one identity forces $a_1$ and $b_1$ to span the same ray, contradicting the distinctness condition (ii) of Definition 2. □

### B. Pseudocontexts of size $k = 2$ and the Bloch representation

The $k=2$ case requires *complex* scalars; the proof that no genuine *real* pseudocontext of size $k=2$ exists is given as Theorem 10(ii) below.

In dimension two the pseudocontext condition admits a complete characterisation in terms of Bloch vectors. This yields a unified and basis-independent description of all pseudocontexts and clarifies the real–complex dichotomy.

Every rank-one projector on $\mathbb{C}^2$ can be written as

$$P_a = \tfrac{1}{2}\big(I + \vec{r}_a \cdot \vec{\sigma}\big), \qquad \vec{r}_a \in \mathbb{R}^3, \quad \|\vec{r}_a\| = 1, \tag{5}$$

where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices. To be specific, for any unit vector $a = \alpha e_1 + \beta e_2 \in \mathbb{C}^2$ in a chosen orthonormal basis $(e_1, e_2)$, the components of the associated Bloch vector $\vec{r}_a = (r_x, r_y, r_z)$ are obtained by computing the expectation values of the Pauli matrices:

$$\begin{aligned} r_x &= \langle a, \sigma_x a\rangle = 2\,\mathrm{Re}(\alpha^*\beta), \\ r_y &= \langle a, \sigma_y a\rangle = 2\,\mathrm{Im}(\alpha^*\beta), \\ r_z &= \langle a, \sigma_z a\rangle = |\alpha|^2 - |\beta|^2. \end{aligned} \tag{6}$$

**Proposition 6** (Bloch representation of pseudocontexts in dimension $d=2$)**.** *Let $(a_1,\dots,a_k)$ and $(b_1,\dots,b_k)$ be $k$-tuples of unit vectors in $\mathbb{C}^2$ with Bloch vectors $\vec{r}_{a_i}, \vec{r}_{b_i} \in \mathbb{R}^3$. Then the following are equivalent:*

1. *$(a_1,\dots,a_k)$ and $(b_1,\dots,b_k)$ form a pseudocontext;*
2. $\sum_{i=1}^{k} P_{a_i} = \sum_{i=1}^{k} P_{b_i}$;
3. $\sum_{i=1}^{k} \vec{r}_{a_i} = \sum_{i=1}^{k} \vec{r}_{b_i}$.

*The common structure operator $T$ has Bloch form*

$$T = \tfrac{k}{2} I + \tfrac{1}{2}\vec{R}\cdot\vec{\sigma}, \qquad \vec{R} = \sum_{i=1}^{k} \vec{r}_{a_i} = \sum_{i=1}^{k} \vec{r}_{b_i}.$$

*Proof.* By Definition 2 and (4), conditions 1 and 2 are equivalent. For the equivalence with 3, substitute the Bloch representation (5) into the structure operator:

$$\sum_{i=1}^{k} P_{a_i} = \sum_{i=1}^{k} \tfrac{1}{2}\big(I + \vec{r}_{a_i}\cdot\vec{\sigma}\big) = \tfrac{k}{2} I + \tfrac{1}{2}\Big(\sum_{i=1}^{k} \vec{r}_{a_i}\Big)\cdot\vec{\sigma}.$$

The identity component $\frac{k}{2} I$ is the same for both tuples, so equality of the two structure operators reduces to equality of the vector sums, giving the stated Bloch form of $T$. □

### C. Explicit $k=2$ example via Bloch vectors

We can now construct an explicit $k=2$ pseudocontext by directly applying the Bloch representation and showing exactly how it arises from non-orthogonal configurations.

**Proposition 7** (Existence of $k=2$ complex pseudocontexts)**.** *There are genuinely nonorthogonal pseudocontexts of size $k=2$ in $\mathbb{C}^n$ for every $n \geq 2$.*

*Proof.* Embed $\mathbb{C}^2$ into $\mathbb{C}^n$ as the span of the first two basis vectors $e_1, e_2$. Define two pairs of unit vectors:

$$u_\pm = \left(\tfrac{\sqrt{3}}{2}, \pm\tfrac{1}{2}, 0, \ldots, 0\right), \qquad v_\pm = \left(\tfrac{\sqrt{3}}{2}, \pm\tfrac{i}{2}, 0, \ldots, 0\right).$$

We obtain their Bloch vectors explicitly using (6). For $u_\pm$, setting $\alpha = \sqrt{3}/2$ and $\beta = \pm 1/2$, we compute:

$$\begin{aligned} r_x &= 2\,\mathrm{Re}\big(\tfrac{\sqrt{3}}{2} \cdot (\pm\tfrac{1}{2})\big) = \pm\tfrac{\sqrt{3}}{2}, \\ r_y &= 2\,\mathrm{Im}\big(\tfrac{\sqrt{3}}{2} \cdot (\pm\tfrac{1}{2})\big) = 0, \\ r_z &= \big|\tfrac{\sqrt{3}}{2}\big|^2 - \big|\pm\tfrac{1}{2}\big|^2 = \tfrac{3}{4} - \tfrac{1}{4} = \tfrac{1}{2}. \end{aligned}$$

Thus $\vec{r}_{u_\pm} = (\pm\frac{\sqrt{3}}{2}, 0, \frac{1}{2})$, which lie in the $xz$-plane of the Bloch sphere.

For $v_\pm$, setting $\alpha = \sqrt{3}/2$ and $\beta = \pm i/2$, we compute:

$$\begin{aligned} r_x &= 2\,\mathrm{Re}\big(\tfrac{\sqrt{3}}{2} \cdot (\mp\tfrac{i}{2})\big) = 0, \\ r_y &= 2\,\mathrm{Im}\big(\tfrac{\sqrt{3}}{2} \cdot (\mp\tfrac{i}{2})\big) = \pm\tfrac{\sqrt{3}}{2}, \\ r_z &= \big|\tfrac{\sqrt{3}}{2}\big|^2 - \big|\pm\tfrac{i}{2}\big|^2 = \tfrac{3}{4} - \tfrac{1}{4} = \tfrac{1}{2}. \end{aligned}$$

Thus $\vec{r}_{v_\pm} = (0, \pm\frac{\sqrt{3}}{2}, \frac{1}{2})$, which lie in the $yz$-plane.

Summing the Bloch vectors within each pair yields the same structure vector:

$$\vec{R} = \vec{r}_{u_+} + \vec{r}_{u_-} = \vec{r}_{v_+} + \vec{r}_{v_-} = (0,0,1).$$

By Proposition 6, this guarantees that the operator sums match in the two-dimensional subspace. Embedding back into $\mathbb{C}^n$ gives, with $V := \mathrm{span}\{e_1, e_2\}$,

$$\begin{aligned} P_{u_+} + P_{u_-} &= P_{v_+} + P_{v_-} = \tfrac{3}{2} P_{e_1} + \tfrac{1}{2} P_{e_2} \\ &= \big(I_V + \tfrac{1}{2}\sigma_z\big) \oplus 0_{n-2} = \begin{pmatrix} 3/2 & 0 \\ 0 & 1/2 \end{pmatrix} \oplus 0_{n-2}. \end{aligned}$$

The four rays are distinct: $u_+$ and $u_-$ differ by a sign in the real second coordinate; $v_+$ and $v_-$ differ by a sign in the imaginary second coordinate; and $u_\pm$ versus $v_\pm$ differ by a factor of $i$ in the second coordinate, which is not a real scalar. Within each pair, $\langle u_+, u_- \rangle = \frac{1}{2} \neq 0$ and $\langle v_+, v_- \rangle = \frac{1}{2} \neq 0$, so neither pair is orthogonal. The pseudocontext is therefore genuine. ☐

*Remark* 8 (Bloch geometry of the complex $k=2$ example). This construction directly illustrates the continuous $U(1)$ freedom present in complex spaces. The pair of Bloch vectors $\{\vec{r}_{v_+}, \vec{r}_{v_-}\}$ is related to $\{\vec{r}_{u_+}, \vec{r}_{u_-}\}$ by a $90^\circ$ rotation about the $z$-axis (which aligns with their common sum $\vec{R}$). Because this rotation preserves $\vec{R}$, it produces a distinct, valid decomposition of the same fixed structure operator $T$.

### D. Classification in $\mathbb{C}^2$

Pseudocontexts in dimension two correspond exactly to distinct decompositions of a fixed vector $\vec{R} \in \mathbb{R}^3$ into $k$ unit vectors.

**Theorem 9** (Complete characterisation in $\mathbb{C}^2$)**.** *Let $k \geq 1$.*

*(i) For $k=1$, no pseudocontext exists.*

*(ii) For $k=2$, pseudocontexts exist and form continuous families.*

*(iii) For all $k \geq 3$, pseudocontexts exist; if $\vec{R} = 0$, they arise from zero-centroid configurations (e.g. regular polygons).*

*Proof.* Part (i) is Proposition 5.

For (ii), fix $\vec{R} \in \mathbb{R}^3$ with $0 < \|\vec{R}\| < 2$. The set of pairs $\{\vec{r}_1, \vec{r}_2\}$ of unit vectors satisfying $\vec{r}_1 + \vec{r}_2 = \vec{R}$ is parametrised by a circle: writing $\vec{r}_1 = \vec{R}/2 + \vec{w}$ and $\vec{r}_2 = \vec{R}/2 - \vec{w}$, the unit-norm condition $\|\vec{r}_i\| = 1$ becomes $\|\vec{w}\|^2 = 1 - \|\vec{R}\|^2/4$ together with $\vec{w} \perp \vec{R}$, so $\vec{w}$ ranges over a circle of radius $\sqrt{1 - \|\vec{R}\|^2/4}$ in the plane perpendicular to $\vec{R}$ through the origin. Distinct positions $\vec{w} \neq \pm\vec{w}'$ on this circle yield distinct unordered pairs $\{\vec{r}_1, \vec{r}_2\} \neq \{\vec{r}_1', \vec{r}_2'\}$, and hence (by Proposition 6) two-tuples that form a pseudocontext.

For (iii), take $\vec{R} = 0$. In $\mathbb{R}^3$ there are infinitely many distinct triples of unit vectors with zero sum, e.g. the vertices of any regular triangle centred at the origin, or any rotated version thereof. By Proposition 6, any two such distinct triples yield a pseudocontext of size $k=3$. ☐

### E. Real–complex dichotomy in two dimensions

**Theorem 10** (Real vs. complex in $d=2$)**.**

*(i) In $\mathbb{C}^2$, genuine pseudocontexts of size $k=2$ exist.*

*(ii) In $\mathbb{R}^2$, no genuine pseudocontext of size $k=2$ exists.*

*(iii) In $\mathbb{R}^2$, genuine pseudocontexts exist for all $k \geq 3$.*

*Proof.* In $\mathbb{C}^2$, rotations about a fixed Bloch vector $\vec{R} \neq 0$ form a continuous $U(1)$ symmetry, yielding inequivalent decompositions; this is the content of Theorem 9(ii).

In $\mathbb{R}^2$, Bloch vectors lie on a circle. For $k=2$, the condition $\vec{r}_1 + \vec{r}_2 = \vec{R}$ uniquely determines the pair up to permutation, so no second decomposition exists. If $\vec{R} = 0$, the vectors are orthogonal and form a context.

For $k \geq 3$, distinct planar configurations with identical vector sum exist (e.g. rotated regular polygons). ☐

*Remark* 11 (Earlier observation). The nonexistence of genuine real pseudocontexts of size $k=2$ in dimension two was observed in [4] without noticing that the situation is different in complex spaces. The Bloch picture above makes the real–complex dichotomy transparent: over $\mathbb{R}$ a decomposition of a fixed structure operator into two rank-one terms is unique up to permutation, whereas over $\mathbb{C}$ a continuous phase degree of freedom produces inequivalent decompositions.

### F. Spectral form

**Proposition 12** (Spectrum in $d=2$)**.** *The eigenvalues of $T$ are*

$$\lambda_{\pm} = \tfrac{k}{2} \pm \tfrac{1}{2}\|\vec{R}\|.$$

*Remark* 13. The vector $\vec{R}$ completely parametrises pseudocontexts in dimension two.

## V. PSEUDOCONTEXTS OF EVERY SIZE $k \geq 2$

The $k=2$ case is best understood via the Bloch-sphere parametrisation; the general $k$-construction below is its root-of-unity extension. The construction below uses a continuous phase $e^{i\phi}$ and is therefore intrinsically complex; the corresponding existence statement over $\mathbb{R}$ for $k \geq 3$ is covered separately by Example 16 in Section VI.

**Theorem 14** (Complex-phase construction)**.** *Let $H$ contain an orthonormal pair $e_1, e_2$. Fix an integer $k \geq 2$, let $\omega = e^{2\pi i/k}$, choose $\alpha \in (0, \pi/2)$, and choose $\phi \in \mathbb{R}$ such that $e^{i\phi}$ is not a kth root of unity. Define unit vectors*

$$\begin{aligned} a_j &= \cos\alpha\, e_1 + \sin\alpha\, \omega^j e_2, \\ b_j &= \cos\alpha\, e_1 + \sin\alpha\, e^{i\phi}\omega^j e_2, \qquad j = 0,\dots,k-1. \end{aligned}$$

*Then the $2k$ rays spanned by the $a_j$ and $b_j$ are pairwise distinct, and*

$$\sum_{j=0}^{k-1} P_{a_j} = \sum_{j=0}^{k-1} P_{b_j} = k\big(\cos^2\alpha\, P_{e_1} + \sin^2\alpha\, P_{e_2}\big).$$

*Hence $(a_0,\dots,a_{k-1})$ and $(b_0,\dots,b_{k-1})$ form a pseudocontext of size $k$. If $\alpha \neq \pi/4$, then the pseudocontext is genuine.*

*Proof.* Write $c = \cos\alpha$ and $s = \sin\alpha$. For each $j$,

$$\begin{aligned} P_{a_j} &= c^2 P_{e_1} + s^2 P_{e_2} + cs\big(\omega^{-j}|e_1\rangle\langle e_2| + \omega^j |e_2\rangle\langle e_1|\big), \\ P_{b_j} &= c^2 P_{e_1} + s^2 P_{e_2} + cs\big(e^{-i\phi}\omega^{-j}|e_1\rangle\langle e_2| + e^{i\phi}\omega^j |e_2\rangle\langle e_1|\big). \end{aligned}$$

Summing over $j = 0,\dots,k-1$ and using $\sum_{j=0}^{k-1}\omega^j = 0$ cancels all off-diagonal terms, giving

$$\sum_{j=0}^{k-1} P_{a_j} = \sum_{j=0}^{k-1} P_{b_j} = k\big(c^2 P_{e_1} + s^2 P_{e_2}\big).$$

*Distinctness.* If $a_j = \lambda a_\ell$ for some scalar $\lambda$, then comparing first coordinates (which equal $c \neq 0$) gives $\lambda = 1$, and then $\omega^j = \omega^\ell$, so $j = \ell$. Hence the $a_j$ are pairwise distinct rays; the identical argument applies to the $b_j$. If $a_j = \lambda b_\ell$, then $\lambda = 1$ and $\omega^j = e^{i\phi}\omega^\ell$, i.e. $e^{i\phi} = \omega^{j-\ell}$, contradicting the assumption that $e^{i\phi}$ is not a $k$th root of unity. Hence no $a_j$ coincides with any $b_\ell$.

*Genuineness.* For $j \neq \ell$ within the $a$-tuple,

$$\langle a_j, a_\ell\rangle = c^2 + s^2\omega^{\ell-j}.$$

This vanishes only if $c^2 = s^2$ (i.e. $\alpha = \pi/4$) and $\omega^{\ell-j} = -1$. So for $\alpha \neq \pi/4$, no two distinct vectors in the $a$-tuple are orthogonal. An identical computation gives $\langle b_j, b_\ell\rangle = c^2 + s^2\omega^{\ell-j}$, so the same conclusion holds for the $b$-tuple. Hence neither tuple is a context when $\alpha \neq \pi/4$. □

*Remark* 15. At the special value $\alpha = \pi/4$, the structure operator becomes $\frac{k}{2} I_{\mathrm{span}\{e_1,e_2\}}$, so the construction yields a unit-norm tight frame on the two-dimensional support. For $k=2$ this reduces to a context; for $k \geq 3$ it gives a highly symmetric pseudocontext (cf. Remark 17). In dimension $d=2$, this construction reduces to the Bloch-vector description of Sec. IV B.

## VI. A SYMMETRIC $k=3$ EXAMPLE

The case $k=3$ admits a very transparent planar representation; it is the finite-dimensional analogue of the Mercedes–Benz frame.

**Example 16** (Mercedes–Benz type pseudocontext)**.** Fix a two-dimensional subspace $V \subset H$ with orthonormal basis $(e_1, e_2)$, and fix an angle $\theta$ satisfying $\theta \not\equiv 0$, $\theta \not\equiv \pi/3$, and $\theta \not\equiv 2\pi/3$ modulo $\pi$. Put, for $j = 0,1,2$,

$$a_j = \cos\Big(\tfrac{2\pi j}{3}\Big)e_1 + \sin\Big(\tfrac{2\pi j}{3}\Big)e_2,$$

$$b_j = \cos\Big(\tfrac{2\pi j}{3} + \theta\Big)e_1 + \sin\Big(\tfrac{2\pi j}{3} + \theta\Big)e_2.$$

Then the six rays are pairwise distinct, and

$$\sum_{j=0}^{2} P_{a_j} = \sum_{j=0}^{2} P_{b_j} = \tfrac{3}{2} I_V.$$

Thus $(a_0,a_1,a_2)$ and $(b_0,b_1,b_2)$ form a genuine pseudocontext of size $k=3$ over $\mathbb{R}$.

*Proof.* In the basis $(e_1,e_2)$, each projector has the form

$$P_{\cos\varphi\, e_1 + \sin\varphi\, e_2} = \begin{pmatrix} \cos^2\varphi & \cos\varphi\sin\varphi \\ \cos\varphi\sin\varphi & \sin^2\varphi \end{pmatrix}.$$

For $\varphi_j = 2\pi j/3$ and for $\varphi_j = \theta + 2\pi j/3$, the trigonometric identities

$$\sum_{j=0}^{2}\cos(2\varphi_j) = 0, \qquad \sum_{j=0}^{2}\sin(2\varphi_j) = 0,$$

which hold for any arithmetic progression of three angles spaced $2\pi/3$ apart, imply that the off-diagonal terms cancel while the diagonal terms each sum to $3/2$. Hence both sums equal $\frac{3}{2}I_V$.

The three excluded residue classes modulo $\pi$ account for all collisions between the two triples:

- $\theta \equiv 0 \pmod{\pi}$: every $b_j$ coincides with $a_j$ or $-a_j$, giving the same ray.

- $\theta \equiv \pi/3 \pmod{\pi}$: the $b$-triple is a cyclic permutation of the $a$-triple as a set of rays.
- $\theta \equiv 2\pi/3 \pmod{\pi}$: the $b$-triple is again a permutation of the $a$-triple.

For all other values of $\theta$, the six rays are pairwise distinct. Genuineness holds because for generic $\theta$ no two vectors within the same triple are orthogonal (the angle between consecutive vectors in each triple is $2\pi/3 \neq \pi/2$), so neither triple is a context. □

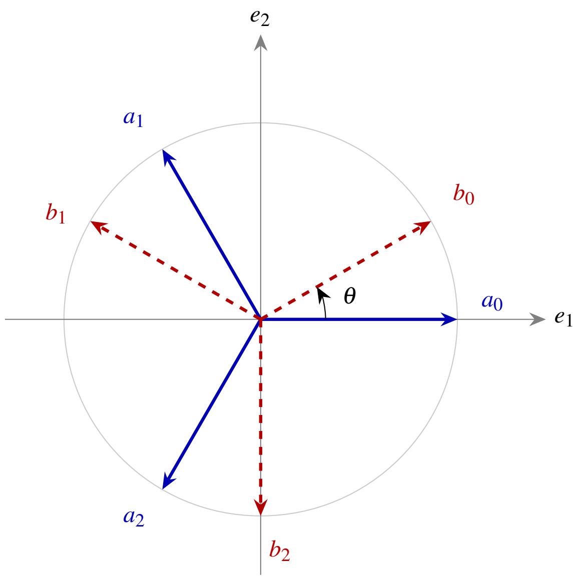


FIG. 1. The Mercedes–Benz pseudocontext of Example 16, drawn for $\theta = \pi/6$. The solid blue rays $a_0, a_1, a_2$ and the dashed red rays $b_0, b_1, b_2$ are two distinct unordered triples in the plane $V$, yet the associated rank-one projectors have the same sum $\frac{3}{2} I_V$.

*Remark* 17 (Complex trine description). Example 16 has a compact complex analogue. Let $\omega = e^{2\pi i/3}$ and embed $V$ into $\mathbb{C}^n$ as $\operatorname{span}\{e_1, e_2\}$. The *trines*

$$u_j = \tfrac{1}{\sqrt{2}}\big(1, \omega^j, 0, \ldots, 0\big),\ v_j = \tfrac{1}{\sqrt{2}}\big(1, e^{i\phi}\omega^j, 0, \ldots, 0\big),$$

with $j = 0, 1, 2$, yield another rank-one decomposition of the same structure operator $\frac{3}{2} I_V$ for any $\phi \in \mathbb{R}$ that is not a multiple of $2\pi/3$. The identity $1 + \omega + \omega^2 = 0$ makes the off-diagonal entries of $\sum_j P_{u_j}$ and $\sum_j P_{v_j}$ cancel, giving the common structure operator $\frac{3}{2} I_V$. This form is most natural in frame-theoretic discussions and in connection with the Weyl–Heisenberg group.

*Remark* 18 (Minimal ambient dimension). Every pseudocontext of size $k \geq 2$ can be realised inside a two-dimensional subspace, and hence embedded into $\mathbb{C}^n$ for any $n \geq 2$: all constructions in Sections IV B and V live in $\operatorname{span}\{e_1, e_2\}$. For $k \geq 3$ the construction also works over $\mathbb{R}^n$. No pseudocontext of any size exists in dimension $n = 1$, since there every unit vector spans the unique ray, making the distinctness condition $|\{a_i\} \cup \{b_i\}| = 2k$ impossible.

*Remark* 19 (Nondegenerate structure operators). The trine family is highly symmetric: its structure operator is a scalar multiple of the identity on its support. More complicated examples with nondegenerate structure operators occur already in dimension three. The main difference is that the pseudocontext in Example 16 has only one constant value in the probability sum, while the advanced constructions admit a larger range of values, still respecting the equality. Explicit three-dimensional implementations of nondegenerate structure operators were derived in [5].

- **15-atom hypergraph ($T_1$).** A pseudocontext of size $k = 3$ yields $\lambda_1 = 2$, $\lambda_2 = \frac{7+\sqrt{21}}{14} \approx 0.827$, $\lambda_3 = \frac{7-\sqrt{21}}{14} \approx 0.173$, with $\operatorname{tr}(T_1) = 3$ and $\lambda_{\max} = 2 > 1$, consistent with Proposition 24.
- **36-atom hypergraph ($T_2$).** A larger true-implies-false gadget gives $T_2 = \operatorname{diag}(1/5, 1/5, 13/5)$ with $\operatorname{tr}(T_2) = 3$ and $\lambda_{\max} = 13/5 > 1$.

Such examples are useful when one wants to move beyond purely symmetric frame orbits.

## VII. SPECTRAL STRUCTURE AND CANONICAL FORMS

In dimension $d = 2$, all spectral properties reduce to the Bloch form of Sec. IV B. We now analyse the general properties of the structure operator $T = \sum_{i=1}^{k} P_{a_i}$ associated with a $k$-tuple of unit vectors $(a_1, \ldots, a_k)$ in $H$.

### A. Gram representation and spectral equivalence

**Definition 20** (Analysis operator and Gram matrix)**.** Let $A\colon \mathbb{C}^k \to H$ be defined by $Ae_i = a_i$, where $(e_i)$ is the standard basis of $\mathbb{C}^k$. The *analysis operator* is $A^\dagger\colon H \to \mathbb{C}^k$, and the *Gram matrix* is

$$G := A^\dagger A = \big(\langle a_i, a_j\rangle\big)_{i,j=1}^{k}.$$

**Proposition 21** (Frame operator factorisation)**.** *The structure operator admits the factorisation $T = AA^\dagger$, and $G = A^\dagger A$.*

**Proposition 22** (Spectral equivalence)**.** *The nonzero spectra of $T$ and $G$ coincide (including multiplicities):*

$$\sigma(T) \setminus \{0\} = \sigma(G) \setminus \{0\}.$$

*In particular,* $\operatorname{rank}(T) = \operatorname{rank}(G) \leq \min(k, n)$.

*Proof.* This is the standard singular-value correspondence between $AA^\dagger$ and $A^\dagger A$: if $AA^\dagger v = \lambda v$ with $\lambda \neq 0$, then $A^\dagger A(A^\dagger v) = \lambda(A^\dagger v)$ and $A^\dagger v \neq 0$. □

### B. Spectral constraints

**Proposition 23** (Trace identities)**.** *The structure operator satisfies*

$$\operatorname{tr}(T) = k, \qquad \operatorname{tr}(T^2) = \sum_{i,j=1}^{k} |\langle a_i, a_j\rangle|^2.$$

*The diagonal terms $i = j$ each contribute 1, giving $\operatorname{tr}(T^2) \geq k$, with equality if and only if the vectors are mutually orthogonal.*

*Proof.* $\operatorname{tr}(T) = \sum_i \operatorname{tr}(P_{a_i}) = k$. For the second identity, $\operatorname{tr}(T^2) = \sum_{i,j} \operatorname{tr}(P_{a_i} P_{a_j}) = \sum_{i,j} |\langle a_i, a_j\rangle|^2$. Diagonal terms contribute $k$; off-diagonal terms are non-negative, and vanish simultaneously if and only if all off-diagonal inner products are zero. □

**Proposition 24** (Non-orthogonality and spectral spread)**.** *If the vectors $(a_i)$ are not mutually orthogonal, then $\operatorname{tr}(T^2) > k$, the spectrum of $T$ is non-flat, and $\lambda_{\max}(T) > 1$.*

**Proposition 25** (Majorisation constraint)**.** *Suppose $k \leq n$. Let $\lambda(T) = (\lambda_1, \ldots, \lambda_n)$ be the eigenvalues of $T$ in decreasing order (padded with zeros to length $n$), and let $d = (1, \ldots, 1, 0, \ldots, 0) \in \mathbb{R}^n$ have $k$ ones and $n-k$ zeros. Then $\lambda(T) \succ d$.*

*Proof.* The diagonal entries of $G$ are $G_{ii} = \|a_i\|^2 = 1$, so the diagonal of $G$ is $(1, \ldots, 1) \in \mathbb{R}^k$. By the Schur–Horn theorem, $\lambda(G) \succ (1, \ldots, 1)$. Since $G$ and $T$ share the same nonzero eigenvalues (Proposition 22), padding $\lambda(G)$ with $n-k$ zeros gives $\lambda(T)$, and $\lambda(T) \succ d$ follows.

*Overcomplete case $k > n$.* When $k > n$ the Gram matrix $G$ is $k \times k$ with rank at most $n$; the majorisation is then best stated for the $k$-dimensional eigenvalue vector of $G$ against $(1, \ldots, 1) \in \mathbb{R}^k$, and the Schur–Horn argument applies to $G$ directly. □

**Proposition 26** (Operator bounds)**.** *The structure operator satisfies $0 \leq T \leq kI$. Moreover, if $\mu := \max_{i \neq j} |\langle a_i, a_j\rangle|$, then*

$$\lambda_{\max}(T) \leq 1 + (k-1)\mu.$$

*Proof.* Positivity $T \geq 0$ is immediate, and $T \leq kI$ since each $P_{a_i} \leq I$. For the eigenvalue bound, apply the Geršgorin circle theorem to $G$: since $G_{ii} = 1$ and $|G_{ij}| \leq \mu$ for $i \neq j$, every eigenvalue of $G$ lies in a disk centred at 1 with radius at most $(k-1)\mu$. Since the nonzero eigenvalues of $T$ and $G$ coincide, $\lambda_{\max}(T) \leq 1 + (k-1)\mu$. □

### C. Commutant and symmetry

**Definition 27** (Commutant)**.** The *unitary commutant* of $T$ is $\mathscr{C}(T) := \{U \in U(H) : UT = TU\}$.

**Proposition 28** (Structure of the commutant)**.** *Let $T = \sum_\alpha \lambda_\alpha P_\alpha$ be the spectral decomposition of $T$ with distinct eigenvalues $\lambda_\alpha$ and orthogonal projections $P_\alpha$ of ranks $m_\alpha$. Then $\mathscr{C}(T) \cong \prod_\alpha U(m_\alpha)$.*

*Remark* 29. Spectral degeneracies ($m_\alpha > 1$) give rise to continuous families of distinct rank-one decompositions of $T$ (as in tight frames), whereas a simple spectrum leads to strong rigidity. This explains why the tight-frame case (Remark 15) is so rich in pseudocontexts.

### D. Canonical normal form

**Proposition 30** (Spectral normal form)**.** *Every structure operator is unitarily equivalent to $\operatorname{diag}(\lambda_1, \ldots, \lambda_r, 0, \ldots, 0)$ with $\lambda_i > 0$, $\sum_{i=1}^r \lambda_i = k$, and $r = \operatorname{rank}(T)$.*

### E. Canonical parametrisation of all decompositions

**Theorem 31** (Canonical decomposition)**.** *Let $T$ be a positive operator of rank $r$ with spectral decomposition $T = \sum_{i=1}^r \lambda_i |e_i\rangle\langle e_i|$, $\lambda_i > 0$, and set $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_r)$. Assume $k \geq r$ (this is necessary, since $\operatorname{rank}(CC^\dagger) \leq k$).*

*Every rank-one decomposition $T = \sum_{j=1}^k P_{a_j}$ with $\|a_j\| = 1$ corresponds to a matrix $C \in \mathbb{C}^{r \times k}$ satisfying*

$$CC^\dagger = \Lambda, \qquad \|c_j\|^2 = 1 \text{ for all } j, \tag{7}$$

*where $c_j$ is the $j$-th column of $C$, via $a_j = \sum_{i=1}^r c_{ij} e_i$. Conversely, every such $C$ defines a valid decomposition.*

*Proof.* Let $T = AA^\dagger$ with columns $a_j$. In the eigenbasis of $T$, write $A = UC$ where $U$ is the unitary embedding the eigenbasis of $T$ on its support into $H$. Then $CC^\dagger = U^\dagger T U\big|_{\operatorname{supp}(T)} = \Lambda$, and $\|c_j\| = \|a_j\| = 1$. The converse is a direct computation. □

**Corollary 32** (Unitary freedom)**.** *Among all rank-one decompositions $T = \sum_{j=1}^k |a_j\rangle\langle a_j|$ into $k$* not necessarily *unit-norm vectors, any two are related by $A \mapsto AU$ for some $U \in U(k)$ (this is the standard $U(k)$-orbit on the affine fibre $\{A : AA^\dagger = T\}$). The unit-norm constraint $\|c_j\| = 1$ in (7) cuts out a real-codimension-$k$ subvariety of this orbit, and what is relevant for pseudocontexts is the quotient by phases acting on individual columns and by permutations of columns, since rays—not vectors—are the data.*

### F. Characterisation of pseudocontexts

**Proposition 33** (Operator equivalence criterion)**.** *Two $k$-tuples $(a_1, \ldots, a_k)$ and $(b_1, \ldots, b_k)$ of distinct rays form a pseudocontext if and only if their matrices $C_A, C_B$ in the eigenbasis of $T$ both satisfy (7):*

$$C_A C_A^\dagger = C_B C_B^\dagger = \Lambda.$$

*Remark* 34. Pseudocontexts correspond to distinct points in the space $\{C \in \mathbb{C}^{r \times k} : CC^\dagger = \Lambda, \|c_j\| = 1\}/\sim$, where $\sim$ identifies decompositions that yield the same set of rays.

### G. Extremal spectral bounds

**Proposition 35** (Rayleigh bounds)**.** *For any density operator $\rho$,*

$$\lambda_{\min}(T) \leq \operatorname{tr}(\rho T) \leq \lambda_{\max}(T).$$

*Both bounds are tight, achieved by pure states on the corresponding eigenvectors of $T$.*

## VIII. TOPOLOGICAL FORCING VS. VECTORIAL REPRESENTATION: THE PROBLEM OF "LOOSE ENDS"

The discrepancy between the purely geometric validity of the $k=2$ complex pseudocontext and the failure of its orthogonality hypergraph to enforce it highlights a crucial distinction in how pseudocontexts can arise.

One and the same configuration of observables—drawn as a Greechie-type orthogonality hypergraph—can admit very different types of probability assignments. These include classical probabilities (mediated by convex sums of two-valued $\{0,1\}$-states) as well as quantum probabilities (mediated by density operators via the Cauchy–Gleason theorem [6]). However, any valid probability assignment on the hypergraph, regardless of its origin, must satisfy the fundamental completeness condition: the sum of probabilities on any context must equal one (that is, $P(I)=1$).

For certain tightly intertwined hypergraphs, this completeness condition alone is sufficient to structurally force a pseudocontext identity. The proof technique, successfully utilized in [4] and applicable to configurations like the aforementioned octagon, relies on finding two distinct "coverings." By summing the completeness conditions of a specific subset of contexts containing the first pseudocontext, and comparing it to a second subset containing the second pseudocontext, one can algebraically cancel out all intermediate auxiliary vertices. When this topological cancellation is perfect, the equality of the pseudocontext probability sums becomes a rigid, state-independent logical necessity of the hypergraph itself, applying equally to classical and quantum probabilities.

However, this covering technique cannot be applied to the $k=2$ complex pseudocontext formed by $u_\pm$ and $v_\pm$ in Proposition 7. If we attempt to equate the probability sums of the $u$-pair and the $v$-pair using the contexts provided, we obtain:

$$P(u_+)+P(u_-)=2-2P(e_3)-\big(P(w_+)+P(w_-)\big), \quad (8)$$
$$P(v_+)+P(v_-)=2-2P(e_3)-\big(P(z_+)+P(z_-)\big). \quad (9)$$

Subtracting (9) from (8) yields:

$$\begin{aligned} &P(u_+)+P(u_-)-\big(P(v_+)+P(v_-)\big) \\ &=P(z_+)+P(z_-)-\big(P(w_+)+P(w_-)\big). \end{aligned} \quad (10)$$

For the pseudocontext identity to be forced purely by the graph, the right-hand side would need to be forced to zero. But $w_\pm$ and $z_\pm$ are "loose ends": they each belong to exactly one context in the hypergraph. Because they do not intertwine with any other contexts, their probability values are completely unconstrained relative to one another. There is no topological mechanism to force $P(w_+)+P(w_-)=P(z_+)+P(z_-)$.

It is precisely these loose ends that allow the classical $\{0,1\}$-valuation (which assigns 0 to $w_\pm$ and 1 to $z_\pm$) to comfortably violate the pseudocontext equality. The identity $P(u_+)+P(u_-)=P(v_+)+P(v_-)$ holds for these specific quantum observables solely due to the "miracle" of their specific complex vector representation, not due to the logical skeleton of their orthogonalities.

To achieve a structurally forced pseudocontext—one where the identity is dictated purely by the graph—we require tighter bounds and configurations without such loose ends. The vertices must intertwine in such a way that they form closed topological loops, enabling the exact algebraic cancellations utilized in the covering proofs. This motivates the search for, and the study of, more complex, higher-dimensional hypergraphs where pseudocontextuality is embedded directly into the orthomodular logic.

## IX. THE OCTAGON PSEUDOCONTEXT VIA DISTINCT COVERINGS

We can rigorously demonstrate the pseudocontextual nature of a tightened—relative to the previous configuration involving $u_\pm$ and $v_\pm$—octagon hypergraph by employing a proof technique based on mutually distinct "coverings," a method we previously utilized in [4].

The hypergraph consists of 20 vertices and 12 contexts of size three. We label the 16 outer vertices along the perimeter as follows: let $v_0,v_1,\dots,v_7$ be the midpoints of the eight octagon edges, and let $v_{01},v_{12},\dots,v_{70}$ be the eight corners. The perimeter is formed by eight straight-line contexts:

$$E_i=\{v_{i-1,i},\ v_i,\ v_{i,i+1}\} \quad \text{for } i=0,\dots,7 \pmod 8.$$

Notice that adjacent midpoints, such as $v_1$ and $v_2$, are not on the same context; rather, their respective perimeter contexts $E_1$ and $E_2$ intertwine at the shared corner vertex $v_{12}$.

The remaining four inner contexts connect opposite midpoints and pass through the four inner vertices $a_1,a_2$ and $b_1,b_2$:

$$\begin{aligned} D_0&=\{v_0,a_1,v_4\}, \quad D_2=\{v_2,a_2,v_6\}, \\ D_1&=\{v_1,b_2,v_5\}, \quad D_3=\{v_3,b_1,v_7\}. \end{aligned}$$

We aim to prove that the sums of probabilities on the inner pairs match: $P(a_1)+P(a_2)=P(b_1)+P(b_2)$. To do so, we construct two distinct coverings of the hypergraph.

**Covering A** focuses on the first pseudocontext pair $\{a_1,a_2\}$, as depicted in Fig. 2(a). We construct it by taking the inner contexts $D_0,D_2$ along with four alternating perimeter edges:

$$A=\{D_0,D_2,E_1,E_3,E_5,E_7\}.$$

Summing the completeness condition over these six contexts yields a total probability of 6. Let us evaluate the multiplicity of each vertex covered by $A$:

- The inner vertices $a_1,a_2$ are covered exactly once.
- The midpoints $v_0,v_4$ and $v_2,v_6$ are covered exactly once (by $D_0,D_2$).
- The remaining midpoints $v_1,v_3,v_5,v_7$ are covered exactly once (by $E_1,E_3,E_5,E_7$).
- Each of the eight corners $v_{i,i+1}$ is covered exactly once because the selected alternating perimeter contexts perfectly span all corners without overlapping.

Thus, covering A perfectly partitions all 16 outer vertices, covering each exactly once:

$$P(a_1)+P(a_2)+\sum_{i=0}^{7}P(v_i)+\sum_{\text{corners}}P(v_{i,i+1})=6. \tag{11}$$

**Covering B** focuses on the second pseudocontext pair $\{b_1,b_2\}$, depicted in Fig. 2(b). We construct it using the other two inner contexts $D_1,D_3$ alongside the complementary set of four alternating perimeter edges:

$$B=\{D_1,D_3,E_0,E_2,E_4,E_6\}.$$

Summing over these six contexts also yields 6. Evaluating the vertex multiplicities reveals:

- The inner vertices $b_1,b_2$ are covered exactly once.
- The midpoints $v_1,v_5$ and $v_3,v_7$ are covered exactly once (by $D_1,D_3$).
- The remaining midpoints $v_0,v_2,v_4,v_6$ are covered exactly once (by $E_0,E_2,E_4,E_6$).
- The selected perimeter contexts $E_0,E_2,E_4,E_6$ again perfectly span all eight corners exactly once.

Hence, Covering B covers the same set of 16 outer vertices, each exactly once:

$$P(b_1)+P(b_2)+\sum_{i=0}^{7}P(v_i)+\sum_{\text{corners}}P(v_{i,i+1})=6. \tag{12}$$

Because Covering A and Covering B perfectly match in their multiplicities over all 16 outer perimeter vertices, we can directly equate (11) and (12). The perimeter terms completely cancel out, leaving:

$$P(a_1)+P(a_2)=P(b_1)+P(b_2).$$

This rigorously establishes the state-independent probabilistic equivalence between the two pairs.

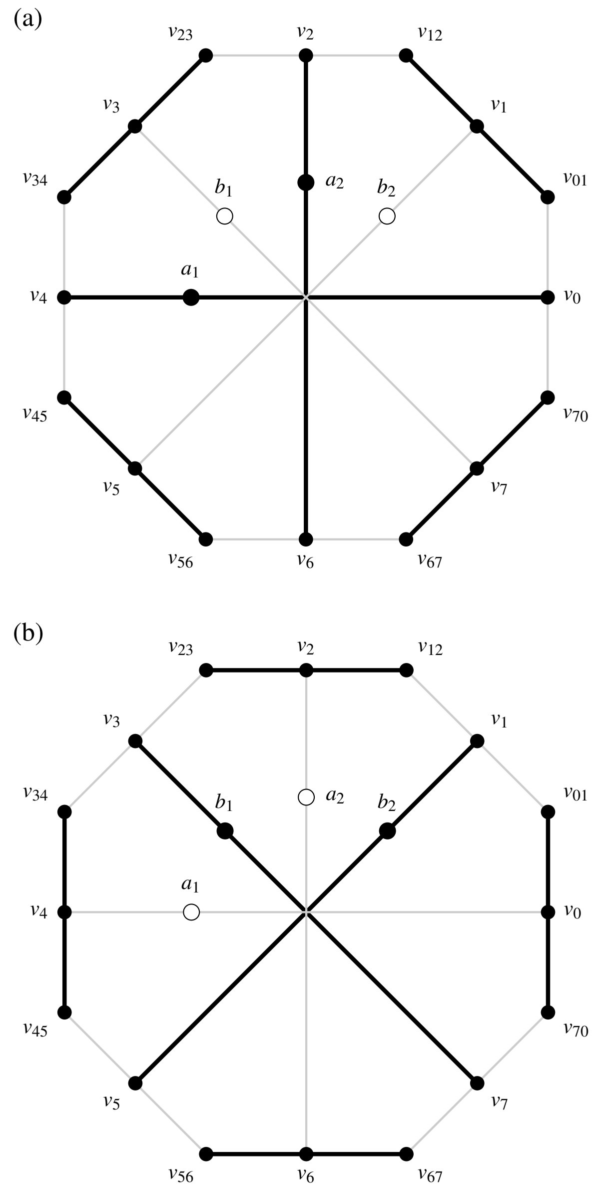


FIG. 2. Graphical representation of the two distinct coverings of the octagon hypergraph. **(a)** Covering A (bold lines) isolates the $(a_1,a_2)$ pseudocontext using inner contexts $D_0,D_2$ and alternating outer edges. **(b)** Covering B (bold lines) isolates the $(b_1,b_2)$ pseudocontext using inner contexts $D_1,D_3$ and the complementary outer edges. Vertices actively engaged in a covering are filled black, while the two strictly uncovered inner vertices in each panel are filled white. Since both coverings exactly partition all 16 outer vertices, their respective probability sums must equate, forcing $P(a_1)+P(a_2)=P(b_1)+P(b_2)$.

### A. The parameter $C$

In the octagon coordinatization—that is, the search for a faithful orthogonal representation [7] of the uniform hypergraph [8] in which vectors assigned to adjacent vertices (or, more generally, to vertices within each hyperedge) are required to be orthogonal—it is convenient to parametrize the inner pair by a single positive parameter $C$. Since only rays matter, all vectors below are understood up to a nonzero scalar multiple.

**Proposition 36** (Meaning of the parameter $C$)**.** *Let $s>0$, set $C:=s^2$, and choose representatives*

$$a_1=(s,1,0), \qquad a_2=(s,-1,0),$$

*with companion rays*

$$b_1=(s,i,0), \qquad b_2=(s,-i,0).$$

*Then the inner pair $\{a_1,a_2\}$ has mutual angle $2\alpha$ satisfying*

$$\cos(2\alpha)=\frac{C-1}{C+1}, \qquad \text{equivalently} \qquad C=\cot^2\alpha.$$

*In particular, $C=1$ is the orthogonal case.*

*Proof.* The normalized representatives of $a_1$ and $a_2$ are

$$\frac{(s,1,0)}{\sqrt{s^2+1}}, \qquad \frac{(s,-1,0)}{\sqrt{s^2+1}}.$$

Their inner product equals $(s^2-1)/(s^2+1)$, which gives the stated formula after substituting $C=s^2$. The same computation applies to the $b$-pair. □

*Remark* 37. The point of the parameter $C$ is that, in the explicit octagon coordinatization used in the main text, the closure conditions depend on the inner pair only through this single scalar.

### B. Exact $\mathbb{C}^3$ coordinates at $C=1$

At $C=1$ the inner pair becomes orthogonal, and the numerical $\mathbb{C}^3$ solution found for the octagon admits a closed-form algebraic realization. As in the main text, the vectors below are ray representatives; normalization is not essential. The configuration is depicted in Fig. 3. A closer inspection in Appendix B 2 shows that this configuration of 21 points in 14 contexts (hyperedges) supports a separating—and thus set representable [9], as discussed in Chapter 3 of Ref. [10]; see Theorem 0 of Ref. [11]—set of 24 two-valued states.

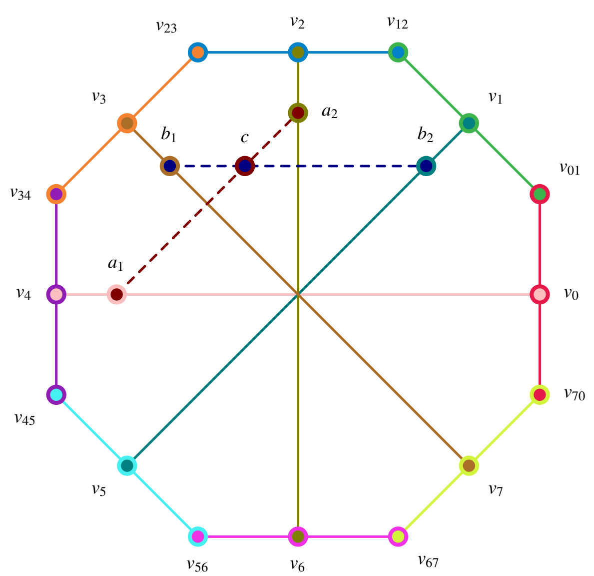


FIG. 3. Redraw of the octagon using exactly the same coordinates as Fig. 2(a)/(b). The 12 original contexts are drawn with solid lines in 12 distinct colors. The two additional $C=1$ contexts are dashed and meet at the intertwining vertex $c$. Concentric disks mark vertices incident with two contexts.

**Example 38** (Exact $\mathbb{C}^3$ realization at $C=1$)**.** Let

$$p=\frac{\sqrt{14}+\sqrt{2}}{4}, \qquad q=\frac{\sqrt{14}-\sqrt{2}}{4}.$$

Then

$$p^2+q^2=2, \qquad pq=\frac{3}{4}.$$

A convenient choice of the inner rays intersection at the additional point $c=(0,0,1)$ is

$$a_1=(1,1,0), \quad a_2=(1,-1,0),$$
$$b_1=(1,i,0), \quad b_2=(1,-i,0).$$

One compatible choice of the eight midpoint rays is

$$v_0=(1,-1,\sqrt{2}), \qquad v_4=(1,-1,-\sqrt{2}),$$
$$v_1=(1,i,p+iq), \qquad v_5=(1,i,-p-iq),$$
$$v_2=(1,1,-i\sqrt{2}), \qquad v_6=(1,1,i\sqrt{2}),$$
$$v_3=(1,-i,q-ip), \quad v_7=(1,-i,-q+ip).$$

One compatible choice of the eight corner rays is

$$v_{01}=\big(-p+i(q+\sqrt{2}),\ \sqrt{2}-p+iq,\ 1-i\big),$$
$$v_{12}=\big(\sqrt{2}-p+iq,\ p-i(q+\sqrt{2}),\ 1+i\big),$$
$$v_{23}=\big(q+\sqrt{2}+ip,\ -q+i(\sqrt{2}-p),\ -1+i\big),$$
$$v_{34}=\big(q+i(p-\sqrt{2}),\ q+\sqrt{2}+ip,\ -1-i\big),$$
$$v_{45}=\big(p-i(q+\sqrt{2}),\ -\sqrt{2}+p-iq,\ 1-i\big),$$
$$v_{56}=\big(-\sqrt{2}+p-iq,\ -p+i(q+\sqrt{2}),\ 1+i\big),$$
$$v_{67}=\big(-q-\sqrt{2}-ip,\ q-i(\sqrt{2}-p),\ -1+i\big),$$
$$v_{70}=\big(-q-i(p-\sqrt{2}),\ -q-\sqrt{2}-ip,\ -1-i\big).$$

A direct symbolic check shows that each listed context is orthogonal. The identities $p^2+q^2=2$ and $pq=\frac{3}{4}$ are sufficient for the simplification. Hence the octagon from Section IX admits an exact $\mathbb{C}^3$ realization at $C=1$.

*Remark* 39. The coordinates above are not unique. Any common unitary transformation, together with arbitrary rescalings of the individual ray representatives, yields an equivalent realization.

*Remark* 40 (Phase symmetry). The exact $\mathbb{C}^3$ realization exploits a complex phase symmetry in the third coordinate. This is one of the reasons why the construction is much more rigid over $\mathbb{R}$ than over $\mathbb{C}$.

### C. No faithful real realization at $C=1$

Example 38 provides an exact realization of the octagon hypergraph in $\mathbb{C}^3$ at the orthogonal limit $C=1$, where the inner pseudocontext splits into two contexts. We show that no faithful realization by real rays exists.

**Proposition 41** (No faithful real realization at $C=1$)**.** *The octagon hypergraph of Section IX, augmented by the two inner $C=1$ contexts intersecting at the vertex $c$, admits no faithful representation by rays in $\mathbb{R}^3$.*

*Proof.* Assume a faithful real realization exists. By a global orthogonal transformation, fix $c=(0,0,1)^T$. All rays sharing a context with $c$ lie in the $xy$-plane. Using rotational freedom in that plane, set

$$a_1=(1,0,0)^T, \qquad a_2=(0,1,0)^T.$$

Let

$$b_1 = (\cos\theta, \sin\theta, 0)^T, \qquad b_2 = (-\sin\theta, \cos\theta, 0)^T,$$

with $\theta \notin \frac{\pi}{2}\mathbb{Z}$ by faithfulness. Define $m = \tan\theta \neq 0$ and $M = 1+m^2 > 1$.

Let the midpoint rays be $v_j = (X_j, Y_j, Z_j)^T$.

*Nonplanarity of midpoint rays.* Every midpoint lies in exactly one inner context, paired with one of the planar reference vectors: $v_0, v_4 \in D_0$ (with $a_1$); $v_2, v_6 \in D_2$ (with $a_2$); $v_1, v_5 \in D_1$ (with $b_2$); $v_3, v_7 \in D_3$ (with $b_1$). Suppose $Z_j = 0$, so that $v_j$ lies in the $xy$-plane. If $v_j$ is paired with $a_1$ (resp. $a_2$), orthogonality forces $X_j = 0$ (resp. $Y_j = 0$), leaving $v_j$ proportional to $a_2$ (resp. $a_1$). If $v_j$ is paired with $b_1$ or $b_2$, the relations $Y = mX$ or $X = -mY$ together with $Z_j = 0$ force coincidence with one of the $b$-rays. In every case this violates faithfulness. Hence $Z_j \neq 0$ for all $j$, and we may normalize

$$v_j = (X_j, Y_j, 1)^T.$$

The inner contexts impose:

$$\begin{aligned}
D_0: &\quad X_0 = X_4 = 0, \qquad Y_0Y_4 + 1 = 0,\\
D_1: &\quad Y_1 = mX_1, \quad Y_5 = mX_5, \quad X_1X_5M + 1 = 0,\\
D_3: &\quad X_3 = -mY_3, \quad X_7 = -mY_7, \quad Y_3Y_7M + 1 = 0.
\end{aligned}$$

Thus

$$X_5 = -\frac{1}{MX_1}, \qquad Y_7 = -\frac{1}{MY_3}.$$

The quantities $X_1$ and $Y_3$ are nonzero: if $X_1 = 0$ (resp. $Y_3 = 0$), then $v_1$ (resp. $v_3$) becomes collinear with a coordinate direction, forcing vertex identifications incompatible with faithfulness.

For perimeter contexts, the condition that adjacent corner rays are orthogonal yields

$$(v_{j-1}\cdot v_j)(v_j \cdot v_{j+1}) - (v_{j-1}\cdot v_{j+1})\|v_j\|^2 = 0.$$

Evaluating at $j = 0$ and $j = 4$ gives

$$Y_0 = \frac{Y_7 + mX_1}{1 - mX_1Y_7}, \qquad Y_4 = \frac{Y_3 + mX_5}{1 - mY_3X_5}, \tag{13}$$

where the denominators are nonvanishing: if $1 - mX_1Y_7 = 0$, the closure equation at $j = 0$ reduces to $Y_7 + mX_1 = 0$, and combining this with $mX_1Y_7 = 1$ yields $-m^2X_1^2 = 1$, which has no real solution. The case $1 - mY_3X_5 = 0$ at $j = 4$ is excluded analogously.

Substituting the inner relations into (13) and simplifying the compound fractions yields

$$Y_0 = \frac{mMX_1Y_3 - 1}{MY_3 + mX_1}, \qquad Y_4 = \frac{MX_1Y_3 - m}{MX_1 + mY_3}.$$

Using $Y_0Y_4 = -1$ from $D_0$, set $u = X_1Y_3$. Expanding the product of these fractions, collecting terms in $u$, and applying the identity $M = 1+m^2$ causes all mixed terms to cancel, yielding the master constraint (see Appendix A)

$$(M^2u^2 + mu + 1) + M(X_1^2 + Y_3^2) = 0. \tag{14}$$

Since $X_1, Y_3 \in \mathbb{R}$,

$$X_1^2 + Y_3^2 \geq 2|X_1Y_3| = 2|u|,$$

and therefore the left-hand side is bounded below by

$$f(u) = M^2u^2 + mu + 1 + 2M|u|.$$

Using $M = 1+m^2$, one has $2M - |m| > 0$, hence

$$M^2u^2 + mu + 2M|u| \geq M^2u^2 + |u|(2M - |m|) \geq 0,$$

with equality only at $u = 0$. Consequently $f(u) \geq 1$ for all $u \in \mathbb{R}$, contradicting the required vanishing. Therefore no real realization exists. □

*Remark* 42 (Minimality of the obstruction). Of the fourteen orthogonality relations defining the $C = 1$ octagon, the proof above invokes only the three inner contexts $D_0$, $D_1$, $D_3$ and the two perimeter closures at $j = 0$ and $j = 4$; the remaining nine relations are not used. The impossibility therefore holds already for this five-relation sub-pattern.

*Remark* 43 (Complex realization). In the complex construction, one has formally $m = i$, hence $M = 1 + i^2 = 0$. The relations above then become singular, and the coercive quadratic structure responsible for the contradiction disappears. Substituting $M = 0$ reduces the constraint to $iu + 1 = 0$, which admits solutions in $\mathbb{C}$.

*Remark* 44. The paper [12] presented the first construction of an orthogonality (Greechie) diagram which can be represented by vectors in complex spaces, but not in real ones. As a by-product of our effort, we found another (and much simpler) such diagram described in this section.

## X. PHYSICAL INTERPRETATION AND CONSTRAINTS

A pseudocontext $(a_1, \ldots, a_k)$, $(b_1, \ldots, b_k)$ implements a state-independent empirical equivalence between two families of rank-one observables $A_i = P_{a_i}$ and $B_i = P_{b_i}$. Although the operators within each family are generally noncommuting, the two linear functionals

$$\rho \longmapsto \sum_i \operatorname{tr}(\rho A_i), \qquad \rho \longmapsto \sum_i \operatorname{tr}(\rho B_i)$$

agree on the whole state space.

The spectral decomposition of $T$ establishes absolute quantum limits on these probability sums. By Proposition 35,

$$\lambda_{\min}(T) \leq \sum_{i=1}^{k} P(a_i) \leq \lambda_{\max}(T). \tag{15}$$

These bounds are tight.

For the 36-atom hypergraph example [5], the structure operator $T_2 = \mathrm{diag}(1/5, 1/5, 13/5)$ confines the quantum probability sum to $[1/5, 13/5] = [0.2, 2.6]$. A classical hidden-variable model based on two-valued ($\{0,1\}$) truth assignments to projections—subject to the consistency requirement that exactly one atom in each orthogonal basis receives the value 1—would allow the sum to range over $\{0,1,2,3\}$, which includes the extreme values 0 and 3. Because $T_2$ has no eigenvalue equal to 0 or 3, quantum mechanics forbids these extremes; the quantum system cannot reach them. This discrepancy is a signature of Kochen–Specker-type contextuality [11, 13].

This state-independent additive equivalence is related to, but distinct from, the operational notion of contextuality introduced by Spekkens [14], which applies to arbitrary experimental procedures (preparations, transformations, and unsharp measurements) rather than to sharp projective measurements alone. A precise comparison is beyond the scope of this note.

In the language of quantum information, $(A_1,\dots,A_k)$ and $(B_1,\dots,B_k)$ are distinct rank-one decompositions of the same positive operator. When this operator is proportional to the identity on its support, the families form tight frames; after normalisation they yield POVMs.

## XI. GENUINELY THREE-DIMENSIONAL REAL PSEUDOCONTEXTS

The constructions of Sections V–VI produce real pseudocontexts whose rays are confined to a two-dimensional subspace. We now exhibit a complementary mechanism, exploiting the spectral structure developed in Section VII, that yields continuous families of real pseudocontexts whose rays span all of $\mathbb{R}^n$. The starting point is the observation that when the structure operator $T$ is proportional to the identity, the unitary commutant (Proposition 28) is the full orthogonal group, so every spatial rotation produces a new rank-one decomposition of the same operator.

### A. The rotation principle for tight frames

**Proposition 45** (Rotation principle)**.** *Let $(a_1,\dots,a_k)$ be a unit-norm tight frame in $\mathbb{R}^n$, i.e.,*

$$T = \sum_{i=1}^{k} P_{a_i} = \frac{k}{n} I_n.$$

*For every $R \in O(n)$, the rotated tuple $(Ra_1,\dots,Ra_k)$ is again a unit-norm tight frame with the same structure operator $T$. The collision set*

$$S^* = \big\{R \in O(n) : [Ra_j] = [a_i] \text{ for some } i, j\big\}$$

*is a closed proper subvariety of $O(n)$ of codimension at least $n-1$, and hence has Haar measure zero. For every $R \in O(n) \setminus S^*$ the $2k$ rays $[a_1],\dots,[a_k],[Ra_1],\dots,[Ra_k]$ are pairwise distinct, and the tuples $(a_1,\dots,a_k)$ and $(Ra_1,\dots,Ra_k)$ form a pseudocontext of size $k$, genuine whenever $k > n$.*

*Proof.* Conjugation by $R \in O(n)$ fixes any scalar multiple of the identity, so $RTR^T = T$. Since $P_{Ra_i} = RP_{a_i}R^T$, summing gives $\sum_i P_{Ra_i} = RTR^T = T$, proving that the rotated tuple is a unit-norm tight frame with the same structure operator.

For each ordered pair $(i, j)$ and each sign $\varepsilon \in \{\pm 1\}$, the locus $\{R \in O(n) : Ra_j = \varepsilon a_i\}$ is a single coset of the stabiliser $\mathrm{Stab}_{O(n)}(a_j) \cong O(n-1)$, hence a closed submanifold of codimension $n-1$ in $O(n)$. The collision set $S^*$ is the finite union of these submanifolds, hence a closed subvariety of codimension at least $n-1$ and Haar measure zero.

For $R \in O(n) \setminus S^*$, no $Ra_j$ is collinear with any $a_i$, so the $2k$ rays are pairwise distinct. When $k > n$, no $k$-tuple of unit vectors in $\mathbb{R}^n$ can be orthogonal, so neither tuple is a context, and the pseudocontext is genuine. □

*Remark* 46. The genericity condition $R \in O(n) \setminus S^*$ is mild: $S^*$ has Haar measure zero in $O(n)$, so a Haar-random rotation produces a pseudocontext with probability one. The valid parameter space $O(n) \setminus S^*$ is open and dense, and itself a smooth manifold of dimension $\binom{n}{2}$.

*Remark* 47 (Spectral interpretation). Proposition 45 is the maximally degenerate case of the unitary commutant analysis of Section VII: when the spectrum of $T$ collapses to a single eigenvalue, the commutant $\mathscr{C}(T)$ becomes the full orthogonal group, and every rotation gives a fresh rank-one decomposition. This sharpens the informal observation made in Remark 15 for the two-dimensional support of the trine: tight frames are precisely the configurations whose pseudocontextual partners form a continuous orbit under a continuous Lie group of symmetries.

**Corollary 48** (Spectrum of tight pseudocontexts)**.** *Let $n \geq 2$. A genuine pseudocontext with structure operator $T = \lambda I_n$ exists if and only if $\lambda$ is of the form*

$$\lambda = \frac{k}{n}$$

*for some integer $k > n$.*

*Proof.* If $T = \lambda I_n$ arises from a $k$-tuple of unit vectors, then by Proposition 23,

$$\mathrm{tr}(T) = k.$$

Since $\mathrm{tr}(\lambda I_n) = n\lambda$, it follows that $\lambda = k/n$.

Moreover, $\mathrm{rank}(T) = n$, while $\mathrm{rank}(T) \leq k$, hence $k \geq n$. If $k = n$, then Proposition 23 implies that the vectors are mutually orthogonal, so the tuple is a context rather than a genuine pseudocontext. Thus $k > n$ is necessary.

Conversely, for every integer $k \geq n$ there exists a unit-norm tight frame $(a_1,\dots,a_k)$ in $\mathbb{R}^n$ (and hence in $\mathbb{C}^n$) satisfying

$$\sum_{i=1}^{k} P_{a_i} = \frac{k}{n} I_n.$$

For $k > n$, Proposition 45 implies that, for a generic rotation $R \in O(n) \setminus S^*$, the tuple $(Ra_1,\dots,Ra_k)$ yields a distinct decomposition of the same operator. The two tuples therefore form a genuine pseudocontext. □

*Remark* 49 (Discrete spectrum of tight operators). The scalar $\lambda$ is constrained to the discrete set

$$\lambda \in \left\{ \frac{k}{n} : k \in \mathbb{N},\ k > n \right\}.$$

In particular, not every positive real value can occur. For fixed dimension $n$, the admissible values form a discrete subset of $(1,\infty)$; as $n$ varies, these values become dense in $(1,\infty)$.

### B. The tetrahedral pseudocontext

The simplest nontrivial application of Proposition 45 in genuine three-dimensional ambient space is provided by the regular tetrahedron, the unique (up to orthogonal equivalence and relabeling) unit-norm tight frame of size four in $\mathbb{R}^3$.

**Example 50** (Tetrahedral pseudocontext)**.** Let

$$\begin{aligned} a_1 &= (0,0,1), \\ a_2 &= \left(\tfrac{2\sqrt{2}}{3}, 0, -\tfrac{1}{3}\right), \\ a_3 &= \left(-\tfrac{\sqrt{2}}{3}, \tfrac{\sqrt{6}}{3}, -\tfrac{1}{3}\right), \\ a_4 &= \left(-\tfrac{\sqrt{2}}{3}, -\tfrac{\sqrt{6}}{3}, -\tfrac{1}{3}\right) \end{aligned}$$

be the vertices of a regular tetrahedron inscribed in the unit sphere of $\mathbb{R}^3$, satisfying $\langle a_i, a_j\rangle = -1/3$ for $i \neq j$. Choose any $R \in SO(3) \setminus S^*$ (i.e. $R$ is not a rotation that maps any $a_j$ onto an $\pm a_i$), and define

$$b_j := Ra_j, \qquad j = 1,\dots,4.$$

Then the eight rays $[a_1],\dots,[a_4],[b_1],\dots,[b_4]$ are pairwise distinct, and

$$\sum_{j=1}^{4} P_{a_j} = \sum_{j=1}^{4} P_{b_j} = \tfrac{4}{3} I_3,$$

so $(a_1,\dots,a_4)$ and $(b_1,\dots,b_4)$ form a genuine pseudocontext of size $k = 4$ in $\mathbb{R}^3$ whose eight rays span all of $\mathbb{R}^3$.

A concrete instance is provided by the quarter-turn about the $x$-axis,

$$R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \in SO(3),$$

which yields

$$\begin{aligned} b_1 &= (0,-1,0), \\ b_2 &= \left(\tfrac{2\sqrt{2}}{3}, \tfrac{1}{3}, 0\right), \\ b_3 &= \left(-\tfrac{\sqrt{2}}{3}, \tfrac{1}{3}, \tfrac{\sqrt{6}}{3}\right), \\ b_4 &= \left(-\tfrac{\sqrt{2}}{3}, \tfrac{1}{3}, -\tfrac{\sqrt{6}}{3}\right). \end{aligned}$$

A direct check shows that none of the $b_j$ is collinear with any $a_i$, confirming that the eight rays are distinct.

*Proof.* The matrix $A = [a_1\ a_2\ a_3\ a_4]$ satisfies $AA^T = \frac{4}{3}I_3$ by direct computation, so $T = \frac{4}{3}I_3$. Since $R \in O(3)$,

$$\sum_{j=1}^{4} P_{b_j} = R\Big(\sum_{j=1}^{4} P_{a_j}\Big)R^T = R \cdot \tfrac{4}{3} I_3 \cdot R^T = \tfrac{4}{3} I_3,$$

giving the common structure operator. The hypothesis $R \in SO(3) \setminus S^*$ excludes ray collisions, so the eight rays are pairwise distinct. Within the $a$-tuple, the inner products $\langle a_i, a_j\rangle = -1/3 \neq 0$ for $i \neq j$ show that no two distinct vectors are orthogonal, so the tuple is not a context; the $b$-tuple is its image under the orthogonal map $R$ and inherits this property. The pseudocontext is therefore genuine, in agreement with Proposition 45 specialised to $n = 3$, $k = 4$. □

*Remark* 51 (Continuous family). For the tetrahedron, $S^* \cap SO(3)$ is a finite union of 1-dimensional submanifolds (cosets of the circle stabilisers $\mathrm{Stab}_{SO(3)}(a_j) \cong SO(2)$), so $SO(3) \setminus S^*$ is an open dense 3-submanifold of $SO(3)$. The proper rotational symmetry group of the regular tetrahedron is $G_{\text{tet}} \cong A_4 \subset SO(3)$, of order 12. Identifying rotations that produce the same unordered pair of 4-tuples of rays (left action of $G_{\text{tet}}$ on the $a$-tuple, right action on the $b$-tuple, both by relabeling) yields a 3-dimensional moduli space of distinct tetrahedral pseudocontexts. The construction therefore delivers a genuinely 3-parameter family of size-4 real pseudocontexts spanning the whole of $\mathbb{R}^3$.

*Remark* 52 (Rigidity of the tight fibre). The unit-norm tight-frame condition $T = \frac{4}{3}I_3$ cuts out a positive-codimension subvariety of the affine space of $3 \times 4$ real matrices. Within the canonical parametrisation of Theorem 31, the corresponding fibre $CC^T = \frac{4}{3}I_3$ with unit columns is acted on transitively by the 3-dimensional commutant $O(3)$, and the unit-norm slice cuts the orbit down to a 3-manifold of distinct frames. Consequently every solution is a critical point of any smooth penalty functional measuring deviation from tightness, which explains why a naive gradient descent toward $T = \frac{4}{3}I_3$ stalls: the target locus is a flat plateau, not an isolated minimum. The rotation principle bypasses optimisation entirely by parametrising this plateau directly.

*Remark* 53 (Physical interpretation). A four-outcome rank-one POVM in $\mathbb{R}^3$ structured like a tetrahedral measurement apparatus may be rotated rigidly by any $R \in SO(3)$. The rotated apparatus describes a physically distinct set of nonorthogonal measurement directions, yet by Example 50 the sum of single-outcome probabilities coincides on every quantum state. The pseudocontextual identity at $k = 4$ is therefore not an isolated algebraic accident but a robust feature of the tetrahedral frame geometry.

**Conjecture 54** (The tetrahedral pseudocontext is not graph–forced)**.** *Let $(a_1,\dots,a_4)$ and $(b_1,\dots,b_4) = (Ra_1,\dots,Ra_4)$ be the tetrahedral pseudocontext of Example 50, with $R \in SO(3) \setminus S^*$. Then the pseudocontextual identity*

$$\sum_{j=1}^{4} P_{a_j} = \sum_{j=1}^{4} P_{b_j} = \tfrac{4}{3} I_3$$

*is not graph-forced over $\mathbb{R}$: there is no finite orthogonality hypergraph $\Gamma$ over $\mathbb{R}^3$ with vertex set containing $\{[a_1],\dots,[a_4],[b_1],\dots,[b_4]\}$ such that the closure conditions $\sum_{v\in C} P(v) = 1$ for $C$ ranging over the contexts of $\Gamma$ entail $\sum_j P(a_j) = \sum_j P(b_j)$ via a covering argument of the type employed in Section IX, for any choice of $R \in SO(3) \setminus S^*$ outside a measure-zero subset.*

*Remark* 55 (Supporting evidence). The eight rays $[a_1],\dots,[a_4],[b_1],\dots,[b_4]$ carry no pairwise orthogonalities among themselves: within each tuple $\langle a_i, a_j\rangle = \langle b_i, b_j\rangle = -\frac{1}{3} \neq 0$ for $i \neq j$, and the cross-orthogonality locus $\{R \in SO(3) : \langle a_i, Ra_j\rangle = 0 \text{ for some } i, j\}$ is a positive-codimension subvariety of $SO(3)$. The minimal hypergraph extension that could cancel both tuples on a shared set of auxiliaries requires eight extra unit vectors $w_k \propto a_{i(k)} \times b_{j(k)}$ subject to eight cross-Gram product conditions of the form

$$\langle a_i, b_{j_1}\rangle \langle a_i, b_{j_2}\rangle = -\tfrac{1}{3},$$

arising from mutual orthogonality of the auxiliaries inside each context. These conditions are eight algebraic equations on the nine entries of the cross-Gram matrix $G = A^T RA$, while $R \in SO(3)$ contributes only three parameters; generically the system is inconsistent, and the special loci where it is consistent collapse onto $S^*$ itself, violating distinctness. This argument rules out the minimal embedding; the conjecture asserts that the same rigidity persists for arbitrarily complex extensions.

*Remark* 56 (Complex analogue). Over $\mathbb{C}$ the parameter count flips: a rotation $U \in U(3)$ has nine real parameters while the cross-Gram entries $\langle a_i, Ub_j\rangle$ are complex. The dimension argument above does not apply, leaving open whether complex tetrahedral-type pseudocontexts admit hypergraph-forcing extensions in $\mathbb{C}^3$.

### C. The tightness obstruction at $k = 3$

The rotation principle of Proposition 45 cannot be applied at $k = 3$ in $\mathbb{R}^3$, because tightness in this regime collapses to orthogonality.

**Proposition 57.** *Let $(a_1, a_2, a_3)$ be a unit-norm tight frame in $\mathbb{R}^3$, i.e. $T = I_3$. Then $(a_1, a_2, a_3)$ is an orthonormal basis, hence a context.*

*Proof.* By Proposition 23, $\mathrm{tr}(T^2) = k + \sum_{i\neq j} |\langle a_i, a_j\rangle|^2$. With $T = I_3$ and $k = 3$ this gives $3 = 3 + \sum_{i\neq j} |\langle a_i, a_j\rangle|^2$, forcing all off-diagonal Gram entries to vanish. □

**Corollary 58.** *Every genuine real pseudocontext of size $k = 3$ whose rays span all of $\mathbb{R}^3$ has nondegenerate structure operator, and therefore lies outside the rotational orbit construction of Proposition 45.*

Such genuinely three-dimensional, nondegenerate $k = 3$ pseudocontexts do exist: explicit coordinatisations were obtained in [5] (recalled in Remark 19), with structure operators having spectra $\{2, (7+\sqrt{21})/14, (7-\sqrt{21})/14\}$ and $\{1/5, 1/5, 13/5\}$ respectively. These configurations are produced not by acting on a single tight base configuration with a continuous symmetry, but by solving the canonical fibre conditions of Theorem 31 directly with a prescribed nondegenerate spectrum, the off-diagonal Gram data being constrained by an underlying orthogonality hypergraph rather than recoverable from a continuous group orbit.

### D. Larger sizes

For every $k \geq 4$, unit-norm tight frames in $\mathbb{R}^3$ exist in abundance, and Proposition 45 produces a 3-parameter family of size-$k$ pseudocontexts whenever the chosen frame has finite ray-stabiliser. Among the most symmetric examples we have the Platonic frames and their pyramid generalisations:

- $k = 4$: regular tetrahedron, $T = \frac{4}{3} I_3$ (Example 50);
- $k = 5$: a regular square pyramid whose four base vectors are elevated to angle $\arccos(1/\sqrt{6})$ from the apex, $T = \frac{5}{3} I_3$;
- $k = 6$: one representative from each of the six antipodal vertex pairs of a regular icosahedron, $T = 2I_3$;
- $k = 7$: a regular hexagonal pyramid whose six base vectors are elevated to angle $\arccos(\sqrt{2}/3)$ from the apex, $T = \frac{7}{3} I_3$;
- $k = 12$: the twelve vertices of a regular icosahedron, $T = 4I_3$.

While Platonic solids naturally furnish perfectly symmetric (equiangular) frames for $k = 4$ and $k = 6$, the cases of $k = 5$ and $k = 7$ illustrate a fundamental structural phase shift. By Gerzon's bound, the maximum number of vectors in a real equiangular tight frame (ETF) in $\mathbb{R}^n$ is $\frac{1}{2} n(n+1)$, which caps at exactly $k = 6$ for $\mathbb{R}^3$. Furthermore, real ETFs only exist for restricted sizes (namely $k = 3, 4, 6$ in $\mathbb{R}^3$). Consequently, no strictly maximally symmetric configurations can exist for $k = 5$ or $k \geq 7$.

However, to compensate for the loss of perfect symmetry, these non-ETF frames acquire internal degrees of freedom. Finite frame theory dictates that the unit-norm tight-frame variety in $\mathbb{R}^3$ modulo rotations has dimension $2k - 8$. Thus, while $k = 4$ represents a rigid shape, the highly symmetric "square pyramid" of $k = 5$ actually sits within a 2-parameter space of smoothly deforming tight frames (e.g., stretching into rectangular or rhombic pyramids), and the $k = 7$ configuration inhabits a 6-parameter deformation space. Each point in these continuous configuration spaces generates its own 3-parameter rotational family of real pseudocontexts spanning $\mathbb{R}^3$.

*Remark* 59. Together with the planar Mercedes–Benz construction (Example 16) and the nondegenerate $k = 3$ realisations in $\mathbb{R}^3$ of [5] (cf. Remark 19), Example 50 completes a coherent picture of real pseudocontexts in low ambient dimension: every $k \geq 3$ admits genuine real realisations spanning

$\mathbb{R}^3$. Those for $k=3$ are necessarily nondegenerate (Corollary 58), those for $k=4$ and $k=6$ admit highly rigid, rotationally parametrised ETF families, and those for $k \geq 5$ (lacking ETF structure) exist within continuous, multidimensional spaces of internally deforming tight realisations. Numerical continuation confirms that unconstrained, purely vectorial $k=3$ pseudocontexts also exist in continuous deformations (e.g., with generic spectra such as $\lambda \approx \{0.68, 1.16, 1.16\} \approx (1/7)\{5,8,8\}$), though they lack the global rotational symmetry of the $k \geq 4$ tight frames.

**Example 60** (Pyramidal tight frames for $k=5$ and $k=7$)**.** By Gerzon's bound [15, Theorem 3.5], no real equiangular tight frame (ETF) can contain more than $\frac{1}{2}n(n+1)$ vectors, meaning no perfectly symmetric Platonic configuration can exist in $\mathbb{R}^3$ for $k=5$ or $k \geq 7$. Nonetheless, highly symmetric unit-norm tight frames can be constructed via regular pyramids. For $k=5$, a tight frame with $T = \frac{5}{3}I_3$ is formed by a square pyramid with apex $v_1 = (0,0,1)$ and four base vectors at an angle of $\arccos(1/\sqrt{6})$ from the apex:

$$v_{2,3} = \left( \pm\sqrt{\frac{5}{6}}, 0, \frac{1}{\sqrt{6}} \right), \quad v_{4,5} = \left( 0, \pm\sqrt{\frac{5}{6}}, \frac{1}{\sqrt{6}} \right).$$

Similarly, for $k=7$, a tight frame with $T = \frac{7}{3}I_3$ is given by a hexagonal pyramid with apex $v_1 = (0,0,1)$ and six base vectors at an angle of $\arccos(\sqrt{2}/3)$ from the apex:

$$v_{j+2} = \left( \frac{\sqrt{7}}{3}\cos\frac{j\pi}{3}, \frac{\sqrt{7}}{3}\sin\frac{j\pi}{3}, \frac{\sqrt{2}}{3} \right) \quad \text{for } j \in \{0,\dots,5\}.$$

Unlike the rigid ETFs at $k=4$ and $k=6$, parameter-counting reveals that the unit-norm tight-frame variety in $\mathbb{R}^3$ has $2k-8$ internal degrees of freedom (modulo rotations). Thus, these $k=5$ and $k=7$ pyramids represent highly symmetric points within 2-parameter and 6-parameter continuous spaces of deforming tight realisations, respectively.

## XII. OPEN PROBLEMS

*Open Problem* 61 (Classification over $\mathbb{C}^n$). Classify, up to unitary equivalence, all pseudocontexts of size $k$ in $\mathbb{C}^n$. Even for small $(k,n)$, the orbit structure is not yet transparent.

*Open Problem* 62 (Lattice representability). Characterise which orthomodular lattices carrying a pseudocontext-type additive identification admit a faithful embedding into some $L(\mathbb{C}^n)$, sharpening the partial results of [5, 16].

*Open Problem* 63 (Frame-symmetry characterisation in the partially degenerate case). Proposition 45 characterises pseudocontexts of tight frames purely in terms of the action of the maximally degenerate commutant $\mathscr{C}(T) \cong O(\operatorname{rank} T)$. For structure operators with spectrum decomposition $T = \sum_\alpha \lambda_\alpha P_\alpha$ and degeneracies $m_\alpha = \operatorname{rank} P_\alpha$, the commutant is $\prod_\alpha U(m_\alpha)$ (Proposition 28), so any two rank-one decompositions of $T$ in the same orbit of $\mathscr{C}(T)$ form a pseudocontext. The converse—whether every pseudocontext arises from such a commutant action—remains open whenever the spectrum is partially degenerate, i.e. at least one $m_\alpha \geq 2$ but not all eigenvalues coincide.

*Open Problem* 64 (Hypergraph-forced higher-dimensional pseudocontexts). The rotation construction of Section XI produces continuous families of pseudocontexts spanning all of $\mathbb{R}^n$, but the pseudocontextual identity in those families is not forced by any orthogonality hypergraph: it relies on the specific tightness of the structure operator and is detectable only at the operator level. In contrast, in [5, 16] and in Section IX of the present paper, the orthogonality relations themselves induce condition (iii) of Definition 2 for every vector representation fulfilling the orthogonality constraints. The systematic classification of hypergraphs that force a pseudocontext identity, and of the higher-dimensional configurations they admit, remains open. This stratum is disjoint from the one resolved in Section XI: the tetrahedral example is not a graph-forced pseudocontext, since the topological covering technique of Section VIII cannot be applied to it.

*Open Problem* 65 (Classification of minimal real pseudocontexts). Theorem 10 establishes that genuine real pseudocontexts of size $k=2$ do not exist in any dimension, while Example 16 demonstrates their existence for $k=3$ in two-dimensional subspaces and [5] exhibits $\mathbb{R}^3$-spanning instances. Thus $k=3$ is the minimal size for genuine real pseudocontexts. Classify all such minimal ($k=3$) examples up to orthogonal equivalence over $\mathbb{R}$, distinguishing the two-dimensional case (where the structure operator is necessarily a scalar multiple of $I_V$, by the analogue of Proposition 57) from the genuinely three-dimensional case (where, by Corollary 58, the structure operator is necessarily nondegenerate).

*Open Problem* 66 (Moduli of tight real pseudocontexts). For each $k \geq n+1$, describe the moduli space $\mathscr{M}^{\text{tight}}_{k,n} = \{\text{unit-norm tight frames of size } k \text{ in } \mathbb{R}^n\}/O(n) \times S_k$, where $S_k$ acts by permutation. Together with Proposition 45, an explicit description of $\mathscr{M}^{\text{tight}}_{k,n}$ would yield a complete parametrisation of the tight stratum of real $\mathbb{R}^n$-spanning pseudocontexts of size $k$. The case $(k,n) = (4,3)$ reduces to a point (the regular tetrahedron, by Example 50); $(k,n) = (5,3)$ already gives a positive-dimensional moduli space.

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### Appendix A: Derivation of the Master Constraint

We provide a step-by-step verification of the algebraic derivation leading from the perimeter closure conditions to the master constraint (14) used in the proof of Proposition 41.

The starting point is the perimeter closure condition stated in the proof: for three rays $v_{j-1}, v_j, v_{j+1}$ forming a context, adjacent corner rays are orthogonal, yielding

$$(v_{j-1}\cdot v_j)(v_j\cdot v_{j+1}) - (v_{j-1}\cdot v_{j+1})\|v_j\|^2 = 0. \tag{A1}$$

Evaluating (A1) at $j=0$ and $j=4$ produces the relations

$$Y_0 = \frac{Y_7 + mX_1}{1 - mX_1Y_7}, \tag{A2}$$

$$Y_4 = \frac{Y_3 + mX_5}{1 - mY_3X_5}, \tag{A3}$$

as stated in the proof. The inner contexts impose the constraints

$$Y_7 = -\frac{1}{MY_3}, \qquad X_5 = -\frac{1}{MX_1}, \qquad M = 1+m^2, \tag{A4}$$

together with the requirement $Y_0Y_4 = -1$ from context $D_0$.

### Simplifying the fractions

Substituting $Y_7$ from (A4) into (A2):

$$Y_0 = \frac{-\frac{1}{MY_3} + mX_1}{1 - mX_1\left(-\frac{1}{MY_3}\right)} = \frac{mMX_1Y_3 - 1}{MY_3 + mX_1}. \tag{A5}$$

Substituting $X_5$ from (A4) into (A3):

$$Y_4 = \frac{Y_3 + m\left(-\frac{1}{MX_1}\right)}{1 - mY_3\left(-\frac{1}{MX_1}\right)} = \frac{MX_1Y_3 - m}{MX_1 + mY_3}. \tag{A6}$$

### Constructing the product condition

Let $u = X_1Y_3$. Using the constraint $Y_0Y_4 = -1$ from context $D_0$ together with (A5) and (A6):

$$\frac{mMu - 1}{mX_1 + MY_3}\cdot\frac{Mu - m}{MX_1 + mY_3} = -1. \tag{A7}$$

Cross-multiplying (A7) yields

$$(mMu-1)(Mu-m) = -(mX_1 + MY_3)(MX_1 + mY_3). \tag{A8}$$

### Expanding both sides

The left-hand side of (A8) expands as

$$\begin{aligned}\mathrm{LHS} &= mM^2u^2 - m^2Mu - Mu + m\\ &= mM^2u^2 - Mu(m^2+1) + m.\end{aligned} \tag{A9}$$

Since $M = 1+m^2$, the coefficient $m^2+1$ equals $M$, giving

$$\mathrm{LHS} = mM^2u^2 - M^2u + m. \tag{A10}$$

The right-hand side of (A8) expands as

$$\begin{aligned}\mathrm{RHS} &= -\left[mMX_1^2 + m^2X_1Y_3 + M^2X_1Y_3 + mMY_3^2\right]\\ &= -\left[u(M^2+m^2) + mM(X_1^2+Y_3^2)\right].\end{aligned} \tag{A11}$$

### Cancellation and the master constraint

Equating (A10) and (A11):

$$mM^2u^2 - M^2u + m = -u(M^2+m^2) - mM(X_1^2+Y_3^2). \tag{A12}$$

Adding $M^2u$ to both sides, the $-M^2u$ term cancels:

$$mM^2u^2 + m = -m^2u - mM(X_1^2+Y_3^2). \tag{A13}$$

Bringing all terms to the left and dividing by $m \neq 0$ (since $\theta \notin \frac{\pi}{2}\mathbb{Z}$ by faithfulness) yields the master constraint (14) in the proof of Proposition 41.

### Positivity contradiction

Since $X_1, Y_3 \in \mathbb{R}$, the inequality $X_1^2 + Y_3^2 \geq 2|X_1Y_3| = 2|u|$ holds. The left-hand side of (14) therefore satisfies

$$\begin{aligned}&M^2u^2 + mu + 1 + M(X_1^2+Y_3^2)\\ &\geq M^2u^2 + mu + 1 + 2M|u|\\ &= M^2u^2 + |u|(2M - |m|) + 1.\end{aligned} \tag{A14}$$

With $M = 1+m^2 > 1$, we have $2M - |m| = 2 + 2m^2 - |m| > 0$ for all real $m$. Consequently the expression in (A14) is strictly positive for every $u \in \mathbb{R}$, contradicting (14). This completes the algebraic verification of the master constraint and the resulting impossibility of a real realization.

## Appendix B: Set representation of the octagon hypergraph via two-valued states

As described in Section 2.3 of Ref. [17], any finite atomic logic equipped with a separating set of dispersion-free probability measures (two-valued states) can be faithfully represented as a partition logic [9–11]. In this formulation, the set representation is constructed by "inverting" the set of two-valued states: every vertex $v$ (atom) of the hypergraph is mapped to the set of indices of those two-valued states that evaluate to 1 ("true") on it. Explicitly, if $M$ denotes the complete set of two-valued states $s_k$, the set representation of a vertex $v$ is given by

$$p(v) = \{k \mid s_k(v) = 1, k \in M\}. \tag{B1}$$

Under this mapping, every context (orthogonal basis) translates into a disjoint partition of the total state space $M$.

### 1. Representation of the incomplete configuration

An exhaustive enumeration of the two-valued states on the tight pseudocontext configuration introduced in Section IX (the incomplete octagon hypergraph depicted in Fig. 2) reveals exactly 26 such states. Because this set of 26 states mutually separates every pair of vertices, the resulting set representation is faithful.

Let us index the set of two-valued states as $M_{26} = \{1,2,\ldots,26\}$. The full evaluation grid—referred to here as the inverted Travis matrix [10, 18]—is presented in Table I, where the states $s_k$ act as the rows and the vertices as the columns. By taking the vertical union of the evaluated indices where $s_k(v) = 1$, we obtain the subsets representing each vertex. Traversing the 16 perimeter vertices in counterclockwise order starting from $v_{56}$, followed by the 4 inner vertices $a_1, b_1, a_2, b_2$, the explicit mapping is as follows:

$$
\begin{aligned}
p(v_{56}) &= \{1,2,3,4,5,6,7,8,9\},\\
p(v_6) &= \{10,11,12,13,14,15,16,17\},\\
p(v_{67}) &= \{18,19,20,21,22,23,24,25,26\},\\
p(v_7) &= \{1,2,3,4,10,11,12,13\},\\
p(v_{70}) &= \{5,6,7,8,9,14,15,16,17\},\\
p(v_0) &= \{1,2,10,11,18,19,20,21\},\\
p(v_{01}) &= \{3,4,12,13,22,23,24,25,26\},\\
p(v_1) &= \{1,5,6,7,10,14,18,19\},\\
p(v_{12}) &= \{2,8,9,11,15,16,17,20,21\},\\
p(v_2) &= \{1,3,5,6,18,22,23,24\},\\
p(v_{23}) &= \{4,7,10,12,13,14,19,25,26\},\\
p(v_3) &= \{5,8,15,16,18,20,22,23\},\\
p(v_{34}) &= \{1,2,3,6,9,11,17,21,24\},\\
p(v_4) &= \{4,5,7,8,12,15,22,25\},\\
p(v_{45}) &= \{10,13,14,16,18,19,20,23,26\},\\
p(v_5) &= \{11,12,15,17,21,22,24,25\},\\
p(a_1) &= \{3,6,9,13,14,16,17,23,24,26\},\\
p(b_1) &= \{6,7,9,14,17,19,21,24,25,26\},\\
p(a_2) &= \{2,4,7,8,9,19,20,21,25,26\},\\
p(b_2) &= \{2,3,4,8,9,13,16,20,23,26\}.
\end{aligned}
$$

#### 2. Representation of the full configuration

When considering the full context configuration of Fig. 3, which includes the additional intertwining point $c$ and two more contexts ($\{a_1,c,a_2\}$ and $\{b_1,c,b_2\}$), the tighter hypergraph imposes stronger constraints and yields a set of exactly 24 two-valued states.

Let us index this new state space as $M_{24} = \{1,2,\ldots,24\}$. The inverted Travis matrix has dimensions $24 \times 21$, corresponding to the 24 states and the 20 previous vertices plus the new vertex $c$ (see Table II). Extracting the non-zero indices column by column provides the explicit set representation for the full hypergraph:

$$
\begin{aligned}
p(v_{56}) &= \{1,2,3,4,5,6,7,8\},\\
p(v_6) &= \{9,10,11,12,13,14,15,16\},\\
p(v_{67}) &= \{17,18,19,20,21,22,23,24\},\\
p(v_7) &= \{1,2,3,4,9,10,11,12\},\\
p(v_{70}) &= \{5,6,7,8,13,14,15,16\},\\
p(v_0) &= \{1,2,9,10,17,18,19,20\},\\
p(v_{01}) &= \{3,4,11,12,21,22,23,24\},\\
p(v_1) &= \{1,5,6,7,9,13,17,18\},\\
p(v_{12}) &= \{2,8,10,14,15,16,19,20\},\\
p(v_2) &= \{1,3,5,6,17,21,22,23\},\\
p(v_{23}) &= \{4,7,9,11,12,13,18,24\},\\
p(v_3) &= \{5,8,14,15,17,19,21,22\},\\
p(v_{34}) &= \{1,2,3,6,10,16,20,23\},\\
p(v_4) &= \{4,5,7,8,11,14,21,24\},\\
p(v_{45}) &= \{9,12,13,15,17,18,19,22\},\\
p(v_5) &= \{10,11,14,16,20,21,23,24\},\\
p(a_1) &= \{3,6,12,13,15,16,22,23\},\\
p(b_1) &= \{6,7,13,16,18,20,23,24\},\\
p(a_2) &= \{2,4,7,8,18,19,20,24\},\\
p(b_2) &= \{2,3,4,8,12,15,19,22\},\\
p(c) &= \{1,5,9,10,11,14,17,21\}.
\end{aligned}
$$

It is straightforward to verify that every context, including the newly added ones, maps to a valid disjoint partition of the complete state space $M_{24}$. For example, the new context $\{a_1,c,a_2\}$ yields the exact partition $p(a_1)\cup p(c)\cup p(a_2) = M_{24}$. This explicit equivalence confirms that the logical skeleton of the tight pseudocontext is fully robust, supporting a classical, mutually separating set representation of automaton logic alongside its quantum representations.

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#### 3. Faithfulness, Greechie's theorem, and the Kochen–Specker distinction

Faithfulness of the set representations constructed above is not merely an empirical observation derivable from Tables I and II: it is guaranteed a priori by the algebraic structure of the hypergraphs. Greechie's Theorem 3.3.5 of [10] states that every finite orthocomplemented poset satisfies what he calls Condition II, and that under Condition II the Γ-sequence algorithm produces a *deterministic Travis matrix*—a $\{0,1\}$-valued evaluation grid whose columns are pairwise distinct (states separate atoms) and whose rows separate states from one another. Both the incomplete octagon hypergraph of Section IX and its $C = 1$ extension are finite orthocomplemented posets; separation therefore follows from Theorem 3.3.5 alone, without inspecting the tables.

TABLE I. The inverted Travis matrix of the 26 two-valued states for the incomplete octagon configuration. Rows $s_1,\dots,s_{26}$ denote the states, and columns represent the vertices.

| State | $v_{56}$ | $v_6$ | $v_{67}$ | $v_7$ | $v_{70}$ | $v_0$ | $v_{01}$ | $v_1$ | $v_{12}$ | $v_2$ | $v_{23}$ | $v_3$ | $v_{34}$ | $v_4$ | $v_{45}$ | $v_5$ | $a_1$ | $b_1$ | $a_2$ | $b_2$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $s_1$ | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| $s_2$ | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| $s_3$ | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |
| $s_4$ | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 |
| $s_5$ | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| $s_6$ | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |
| $s_7$ | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 |
| $s_8$ | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 |
| $s_9$ | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |
| $s_{10}$ | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| $s_{11}$ | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| $s_{12}$ | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 |
| $s_{13}$ | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 |
| $s_{14}$ | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 0 |
| $s_{15}$ | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 |
| $s_{16}$ | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 |
| $s_{17}$ | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 |
| $s_{18}$ | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| $s_{19}$ | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 0 |
| $s_{20}$ | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 |
| $s_{21}$ | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 1 | 0 |
| $s_{22}$ | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 |
| $s_{23}$ | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 |
| $s_{24}$ | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 |
| $s_{25}$ | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 0 |
| $s_{26}$ | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 1 |

TABLE II. The inverted Travis matrix of the 24 two-valued states for the full octagon configuration. Rows $s_1,\dots,s_{24}$ denote the states, and columns represent the vertices.

| State | $v_{56}$ | $v_6$ | $v_{67}$ | $v_7$ | $v_{70}$ | $v_0$ | $v_{01}$ | $v_1$ | $v_{12}$ | $v_2$ | $v_{23}$ | $v_3$ | $v_{34}$ | $v_4$ | $v_{45}$ | $v_5$ | $a_1$ | $b_1$ | $a_2$ | $b_2$ | $c$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $s_1$ | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| $s_2$ | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 |
| $s_3$ | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 |
| $s_4$ | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 |
| $s_5$ | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| $s_6$ | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 |
| $s_7$ | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |
| $s_8$ | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 |
| $s_9$ | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| $s_{10}$ | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| $s_{11}$ | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| $s_{12}$ | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 |
| $s_{13}$ | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 0 |
| $s_{14}$ | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| $s_{15}$ | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 |
| $s_{16}$ | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 |
| $s_{17}$ | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| $s_{18}$ | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 |
| $s_{19}$ | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 |
| $s_{20}$ | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 0 |
| $s_{21}$ | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| $s_{22}$ | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 |
| $s_{23}$ | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 |
| $s_{24}$ | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 0 | 0 |

This should be contrasted with the situation arising in the Kochen–Specker theorem [11]. There, a two-valued state on the partial Boolean algebra $\mathrm{B}(E^3)$ of one-dimensional subspaces of $\mathbb{R}^3$ must satisfy the additional *global* coloring constraint: exactly one member of every orthogonal triple spanning $\mathbb{R}^3$ receives the value 1. This cross-context requirement is strictly stronger than the purely order-theoretic notion of a two-valued state employed here, where each context is required only to sum to 1 in isolation. Theorem 0 of [11] characterizes imbeddability of a partial Boolean algebra into a

Boolean algebra as equivalent to the existence of a *separating* family of such globally consistent homomorphisms; Theorem 1 then shows, via the 117-point graph $\Gamma_2$, that no such family exists for $\mathrm{B}(E^3)$, and the smaller 17-point graph $\Gamma_3$ already witnesses the failure with two vertices $a, b$ satisfying $h(a) = h(b)$ in every admissible homomorphism. No analogous obstruction arises in the present setting: the completeness condition $\sum_{v \in C} P(v) = 1$ is imposed context-by-context without any requirement of global geometric consistency, and it is precisely this weaker structure that places both octagon configurations firmly within Greechie's framework, where a separating set of two-valued states is always guaranteed to exist.


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